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SUPERSEDING
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MILITARY SPECIFICATION

LABORATORY, PETROLEUM, BASE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type assembly of standard metal cabinets, petroleum testing equipment contained therein, mechanical equipment and supplies, and standard shipping cases for these units.

2. APPLICABLE DOCUMENTS

2.1. The following documents of the issue in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

H-B-173	—Brushes, Cleaning; Laboratory Glassware.
O-C-105	—Calcium Chloride, Dehydrate and Calcium Chloride, Anhydrous; Technical.
O-C-265	—Chemicals, Analytical; General Specification For.
O-E-760	—Ethyl Alcohol (Ethanol); Denatured Alcohol; and Proprietary Solvent.
P-S-661	—Solvent, Drycleaning.
BB-C-101	—Carbon Dioxide (CO_2); Compressed.
BB-F-671	—Freon Gas, Compressed (Chlorofluoro Hydrocarbons).
BB-G-110	—Gases, Fuel, Liquefied Petroleum, (Butane, Propane, Mixtures).
BB-O-925	—Oxygen, Technical, Gas and Liquid.
DD-D-205	—Desiccators; Laboratory, and Desiccator Plates.
DD-F-796	—Funnel, Common, Laboratory, Fluted or Ribbed.
DD-F-806	—Funnel, Common, Laboratory, Smooth.
DD-S-636	—Spot-Plate, White Porcelain.
DD-T-741	—Tubing, Glass (Laboratory Use).
DD-V-581	—Volumetric Apparatus; Glass.
DD-W-131	—Watch Glass, Laboratory.
FF-H-11[illegible]	—Hinges, Hardware, Builders.
FF-W-556	—Wool; Steel.
GG-C-4[illegible]1	—Clamps and Clamp-Holders, Laboratory.

FSC 6640

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GG–C–418 —Clamps; For Rubber Tubing, Laboratory.
GG–R–391 —Rings, Support, Laboratory, Metal.
GG–S–593 —Spatulas, Metal-Blade; Laboratory.
GG–S–764 —Stopwatch, Laboratory.
HH–G–450 —Gloves, Cloth (Asbestos).
QQ–C–576 —Copper Flat Products with Slit, Slit and Edge-Rolled, Sheared, Sawed, or Machined Edges, (Plate, Bar, Sheet, and Strip).
TT–C–490 —Cleaning Methods and Pretreatment of Ferrous Surfaces For Organic Coatings.
TT–E–529 —Enamel, Alkyd, Semi-Gloss.
TT–P–636 —Primer Coating, Synthetic, Wood and Ferrous Metal.
TT–M–261 —Methyl-Ethyl-Ketone (For Use In Organic Coatings.
VV–B–231 —Benzene (Benzol), Technical.
ZZ–G–421 —Gloves, Rubber; Surgeons.
ZZ–S–751 —Stoppers, Bottle, Rubber.
ZZ–T–831 —Tubing; Rubber.
CCC–C–440 —Cloth, Cotton, Cheesecloth.
GGG–C–589 —Cork Pressers, Rotary.
GGG–C–591 —Corkscrews.
GGG–F–325 —File, Hand (American Pattern) and Rasp, Hand.
GGG–G–17 —Gages, General Purpose.
GGG–G–521 —Goggles, Industrial, Rubber Frame.
GGG–O–591 —Oiler, Hand.
GGG–P–791 —Pullers, Nail, Hand.
GGG–S–121 —Screwdriver and Screw Starter, Hand.
GGG–S–278 —Shears and Scissors.
GGG–S–291 —Shears, Metal Cutting; Bench and Hand.
GGG–W–631 —Wrenches, Adjustable; Open-End, Auto and Monkey.
GGG–W–651 —Wrenches; Pipe Wrench and Basin Wrench.
JJJ–C–561 —Cotton, Sterile (Purified and Non-sterile).
LLL–S–731 —Stopper, Bottle, Cork.
NNN–B–175 —Beaker, Laboratory; Glass.
NNN–B–600 —Borer Set, Cork.
NNN–B–795 —Burner, Gas, Laboratory.
NNN–C–680 —Crucibles, Filtering and Ignition; and Covers; Laboratory.
NNN–D–340 —Dish, Evaporating, Nonmetallic.
NNN–F–240 —Flasks, Laboratory.
NNN–F–600 —Frame, Laboratory Apparatus Support.
NNN–F–800 —Funnel, Separatory.
NNN–G–698 —Grid, Wire Gauze, Laboratory.
NNN–M–560 —Motor and Pestle.
NNN–R–350 —Ring, Cork, Laboratory.
NNN–R–560 —Rod, Stirring, Laboratory.
NNN–S–750 —Stand, Laboratory Apparatus Support.
NNN–T–685 —Triangle, Wire, Laboratory.
PPP–T–60 —Tape; Pressure Sensitive Adhesive, Waterproof For Packaging and Sealing.

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MIL–M–191 —Mercury
MIL–B–3473 —Bead, Glass.
MIL–L–7808 —Lubricating Oil, Aircraft Turbine Engine, Synthetic Base.
MIL–L–10547 —Liners, Case, and Sheet, Coverwrap; Water-Vaporproof or Waterproof, Flexible.
MIL–C–11777 —Cylinder, Freon, Methyl Chlorine, and Sulfur Dioxide (ICC–4B and ICC–4BA).
MIL–C–12661 —Cylinder, Compressed Gas, With Valves (ICC–3A or ICC–3AA).
MIL–T–16049 —Tachometer, Electrical, Self-Generating Mechanical, Fixed Mounting and Hand Held, and Viberating Reed.
MIL–T–16644 —Tapes, Measuring, Tank Level Gaging.
MIL–C–20033 —Crucible Holder, Walter.
MIL–F–50026 —Fiber, Asbestos, Filter.

STANDARDS

FEDERAL

Fed. Std. No. 48 —Tolerances for Steel and Iron Wrought Products.
Fed. Std. No. 595 —Colors
Fed. Test Methods Std. No. 791 —Lubricants, Liquid Fuels, and Related Products; Methods of Testing.

MILITARY

MIL–STD–105 —Sampling Procedure and Tables for Inspection by Attributes.
MIL–STD–129 —Marking for Shipment and Storage.
MIL–STD–130 —Identification Marking of U.S. Military Property.
MS 35989 —Stopcock, Laboratory Apparatus, Straight Bore Type.
MS 35990 —Tripod, Laboratory Apparatus, Concentric Ring.
MS 36022 —Tongs, Laboratory, Beaker Type.
MS 36023 —Tongs, Laboratory, Crucible, Double-Bent.
MS 36024 —Tongs, Laboratory, Crucible, Double-Bent, Long.
MS 36069 —Wash Bottle, Laboratory, Insulated, W/Rubber Bulb.
MS 36074 —Bulb, Pressure and Circulating, Laboratory, Rubber.
MS 36095 —Glass Wool, Filtering.
MS 36205 —Scraper, Laboratoy, Diagonally Shaped.
MS 36252 —Corrosion Test Strip.

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DRAWINGS

QUARTERMASTER CORPS

	TITLE	UNIT
5–14–322	Laboratory, Petroleum, Base; Water Bath, Oven and Stool	No. 1
5–14–323	Laboratory, Petroleum, Base; Viscosimeter, High Temperature, Kinematic.	No. 2
5–14–324	Laboratory, Petroleum, Base, Distillation.	No. 3
5–14–325	Laboratory, Petroleum, Base; Distillation.	No. 4
5–14–326	Laboratory, Petroleum, Base; Muffle Furnace and Penetrometer.	No. 5
5–14–327	Laboratory, Petroleum, Base; ASTM Gum.	No. 6
5–14–328	Laboratory, Petroleum, Base; Water Still.	No. 7
5–14–329	Laboratory, Petroleum, Base; Tetraethyl Lead and Sulfur.	No. 8
5–14–330	Laboratory, Petroleum, Base; Storage Cabinet.	No. 9 10, 11 12, 13 14, 24 26
5–14–331	Laboratory, Petroleum, Base; Chemical Storage.	No. 15
5–14–332	Laboratory, Petroleum, Base; Balance.	No. 16
5–14–333	Laboratory, Petroleum, Base; Sink.	No. 17
5–14–334	Laboratory, Petroleum, Base; Oxidation Stability.	No. 18
5–14–335	Laboratory, Petroleum, Base; Fume Hood.	No. 19
5–14–336	Laboratory, Petroleum, Base; Viscosimeter. Low Temperature Kinematic.	No. 20
5–14–337	Laboratory, Petroleum, Base; Refrigerator.	No. 21
5–14–338	Laboratory, Petroleum, Base; Cloud and Pour Point.	No. 22
5–14–339	Laboratory, Petroleum, Base; Compressor.	No. 23
5–14–340	Laboratory, Petroleum, Base; Centifuge and Channel Point.	No. 25
5–14–341	Laboratory, Petroleum, Base; Armor-ply Shipping Container.	
5–14–342	Laboratory, Petroleum, Base; Pipe Rack Details.	
5–14–343	Laboratory, Petroleum, Base; Elevating Platform Mechanism A.	
5–14–344	Laboratory, Petroleum, Base; Elevating Platform Mechanism B.	
5–14–345	Laboratory, Petroleum, Base; Details of Drawer and Door Catch Mechanism.	
5–14–346	Laboratory, Petroleum, Base; Wiring Diagram A.	
5–14–347	Laboratory, Petroleum, Base; Wiring Diagram B.	
5–14–348	Laboratory, Petroleum, Base; Wood Crates.	
5–14–374	Laboratory, Petroleum, Base; Wiring for Pour Point.	No. 22
5–14–375	Laboratory, Petroleum, Base; Typical Construction of Cabinet Body.	

(*Note.* Miniature copies of Drawings 5–14–322, 5–14–335, 5–14–337, 5–14–343 and 5–14–346 identified as figures 1 through 5 are attached for information purposes only.)

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained for the procuring activity or as directed by the contracting officer.)

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2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BOARD OF FIRE UNDERWRITERS
National Electric Code

(Application for copies should be addressed to the National Board of Fire Underwriters, 85 John Street, New York 38, N.Y.; 222 West Adams Street, Chicago 6, Ill.; or Merchants Exchange Building, San Francisco 4, Calif.)

UNITED STATES DEPARTMENT OF COMMERCE
CS 45–60—Douglas-fir Plywood

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

3. REQUIREMENTS

3.1 **Alternate components.** Components offered as equivalent to components specified hereinafter and on the reference drawings as specific manufacturer's part number or equal shall be functionally equal, of equal or better quality, with the manufacturer's part number identified. The incorporation and inclusion of such a component in the design of the specified end product shall not require modification or change to any other specified component, and shall not reduce ease of maintenance to it or any other components, unless such modification or change is specifically approved by the contracting officer. Prior to manufacture of the preproduction sample, the supplier shall submit for the contracting officer's approval a list identifying each proposed alternate component, together with proof that each listed component, complies with requirement specified above. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data, authorizes the commencement of preproduction sample manufacture and does not relieve the supplier of the responsibility that these components perform in accordance with specified requirement when incorporated into the end product.

3.2 **Preproduction sample.** Unless otherwise specified (see 6.2), before production is commenced, a sample cabinet complete with equipment and shipping container shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The purpose of testing the preproduction sample is: (1) to determine whether the item manufactured by the supplier has the performance characteristics required by the specifications; and (2) to evaluate the acceptability of parts and components used under the "or equal" provisions when assembled into a complete end item. It is not the purpose of preproduction sample testing to determine compliance with the detailed requirements of the specification and approval of the preproduction sample does not constitute approval of the item as complying with the detailed requirements of the specifications inasmuch as each production item must comply with the detailed requirements of the specifications. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.3 **Materials.** The material specified shall conform to the applicable specifications and requirements specified hereinafter. Details not specifically covered by this specification may be of manufacturer's normal commercial practice provided they meet the general requirements specified herein. Tolerances for sheet steel shall be in accordance with Standard Fed. Std. No. 48.

3.3.1 *Primer.* The primer shall conform to Specification TT–P–636.

3.3.2 *Enamel.* The enamel shall conform to Specification TT–E–529 and the color shall be semi-gloss olive drab No. X24087 of Standard Fed. Std. No. 595.

3.4 **Military engine.** When specified (see 6.2), the Coordinating Oil Testing (C.O.T.)

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Military engine shall conform to the requirements shown on table III.

3.5 **Equipment, apparatus and supplies.** Equipment, apparatus and supplies for each unit shall be as listed in tables I, III and IV and shall be stored in the individual cabinets or shipping containers as applicable. When specified (see 6.2) items listed in table V shall be included in Air Force procurements only.

3.6 **Chemicals.** When specified (see 6.2 and 6.3), chemicals shall be finished in accordance with table II.

3.7 **Packing list.** The supplier shall furnish a packing list with each unit, listing the unit number, equipment, apparatus and supplies in accordance with table VI.

3.8 **Construction.** Construction details shall be as specified herein and as shown on applicable drawings. The reference of commercial products is not intended to preclude or restrict the furnishing of other products that shall meet the same design and performance characteristics. However, all deviations from this specification shall be brought to the attention of the contracting officer when submitting bids and written approval for such deviations secured before proceeding with construction.

3.8.1 *Cabinets.*

3.8.1.1 *Frames.* Frames shall be die-formed sections, welded together for maximum strength and rigidity. All cabinets shall be reinforced with angle iron frame around the entire perimeter at the bottom of the cabinet as shown on applicable drawings. The base of the cabinets shall be provided with four gusset plates at the corners, flanged on all sides for maximum rigidity, and shall contain leveling bolts as shown on Drawing No. 5–14–345. Provision shall be made so that these leveling bolts are accessible from the interior of the cabinet.

3.8.1.2 *Cabinet exterior.* All cabinet exteriors, except tops, shall be manufactured from cold rolled steel sheets of the following gages:

(a) Sides, backs, bottoms, front, rails, and intermediate rails — .0478 inch (18 gage).

(b) Doors and drawer fronts ____ .0359 inch (20 gage).

(c) Inside door panels and drawer head panels ____ .0299 inch (22 gage).

(d) Drawer slides and case work tracks ____ .0598 inch (16 gage).

All tops shall be fabricated from .0500 inch (18 gage) corrosion-resisting steel, commercial type 302, 18–8, and shall be polished after welding to a No. 4 finish.

3.8.1.3 *Welding.* All welding shall be done by the electric arc process. No gas or carbon arc welding shall be used. All corners and joints of the stainless steel tops shall be welded by the heliarc or some other inert gas-shrouded arc welding process. All welds shall be ground smooth and polished flush with the adjacent material. All welding rods or filler rods shall be of the same composition as the material being welded.

3.8.1.4 *Cabinet construction.* All cabinets and tops shall conform to the dimensions shown on applicable drawings. Each unit shall have attached to the back, piping supports and piping with couplers, as indicated on the drawings. All doors shall close in rabbetted recesses fitted with suitable rubber bumpers. Doors shall be hung on butt type or continuous type semi-concealed, fast pin hinges conforming to Specification FF–H–116. Continuous type hinges shall be securely welded to the cabinet proper behind the upright corner posts. The door shall be reinforced with .1046 inch (12 gage), channel-shaped steel clips securely welded to the interior of the door back at point of fastening to the hinge leaf. The hinge leaf shall be tapped for ¼–20 machine screws. When butt type hinges are used they shall be furnished with backing plates as shown on Drawing 5–14–375. Doors installed with butt type hinges shall show no evidence of

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distortion when tested as specified in 4.3.3. Door drawer fronts shall be double wall, pan-type, construction with all corners electrically welded. Door heads shall be sound-deadened and reinforced with box-type channel section. Drawer bodies shall be constructed of .0359 inch (20 gage) steel, with reinforced Z-shaped tracks spot welded to either side. All drawers shall be equipped to operate on four ball bearing rollers and shall be equipped with positive stops to prevent inadvertent removal. All doors and drawers shall be furnished with a flush type, recessed drop handle with a corrosion-resisting steel, spring-loaded locking mechanism which shall securely lock the door or drawer in a closed position. When door backs are used for holding or fastening mechanical equipment, suitable tapping plates and reinforcing channels shall be provided so that there is no unnecessary strain on either the door panel or the hinges. All nuts, bolts, screws, and any unpainted fasteners shall be plated or of noncorrosive materials.

3.8.1.5 *Drawer fasteners and packing.* Drawer bodies shall be fitted with positive fastening devices to firmly retain in position all apparatus and equipment stored in the respective drawers. Those drawers which are fitted for glassware and delicate instrument storage shall be lined with a resilient and shock-absorbing material, die-cut to completely receive each item of apparatus. The fastening devices and packing material shall be adequate to prevent breakage or displacement of the apparatus regardless of the position in which the cabinets might be placed.

3.8.1.6 *Elevating platforms and mechanisms.* All elevating platforms shall ride in suitable tracks on ball bearing rollers, with a minimum of side or end play. The elevating platform shall be firmly retained by these tracks throughout its entire travel, and a suitable locking mechanism shall be provided so as to lock the platform in either the raised or lowered position. The platforms shall be provided with spring sash balances of correct tension to adequately provide for the items to be mounted on each individual platform. The various pieces of equipment mounted on these platforms shall be so modified as to permit firm and adequate fastening to the platform itself for shipping purposes, and yet be easily removable for replacement.

3.8.1.7 *Fume hood construction.* The frame of fume hoods shall be constructed of 2 by 2 by ⅛ inch angle pieces, welded at all joints so as to form an integral one-piece frame to which shall be fastened all Transite or equal panels and baffles. The hood exhaust blower shall be mounted on a cork insulated platform or hard rubber pad which shall be set in a sealed chamber in the upper section of the hood. The unit shall exhaust through the top of the hood as shown on the applicable drawings, and the blower shall be inclosed in a cast iron housing and shall be capable of moving 855 cubic feet of air per minute. The motor driving the fan shall be a ⅙ H.P. 115 V., 60 cycle, 1725 r.p.m. motor. The sash of the hood shall be glazed with ¼ inch clear Flexiglas and shall be fastened to the counter weights by stainless steel cables which shall ride over bronze pulleys. The hood shall be provided with utilities as shown on the applicable drawings. All nuts, bolts, screws and any unpainted fasteners shall be plated or of a noncorrosive material.

3.8.1.8 *Finish.* Interior and exterior surfaces of the frame and cabinet, except the stainless steel top, shall be prepared for painting in accordance with any method or type of Specification TT–C–490. The prepared surfaces shall then be coated with primer specified in 3.3.1. The primer shall level out to a uniform film, free from runs; or areas of no film. When dry, the primed surfaces shall be coated with enamel specified in 3.3.2. The enamel shall level out to a smooth and uniform film, free from runs, wrinkles, drops, or areas of no film. The total dry film thickness of primer and enamel shall be 1.4 mils minimum.

3.8.1.9 *Service lines and supports.* Service lines are to be mounted on the rear of each cabinet except units number 6 and 11 (base

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for fume hoods) and securely fastened in movable tracks as indicated on the applicable drawings. The tracks shall be so arranged to permit the lowering of all lines behind the cabinets. In use, the lines shall be capable of being raised so as to come above the top of the cabinets and securely locked in position at that point. All lines shall be provided with union connectors so that one unit may be fastened to the next, providing a continuous flow of services to all parts of the laboratory. Flexible metal braided hose of the size and quantity called for in table I shall be provided to permit connection of lines around corners, over doorways, et cetera. These lines shall be carried on the racks as indicated. All piping and hose connectors shall be color coded as follows: gas ——yellow; water——white; air——gray; drain——black.

3.8.1.10 *Electrical.* The assembly of cabinets shall be provided with surface-type panel boards, each of which is to serve six cabinets. Each panel board shall contain a 100-ampere main circuit breaker, rated 140 to 240 volts AC, and eight 1-pole, 30 ampere circuit breakers. Electric wiring shall be carried in 1 inch heavy wall rigid conduit securely mounted on the pipe rack as indicated on the applicable drawings. Connections, junction boxes and wiring shall be in accordance with Drawing 5–14–346. A positive and permanent identification shall be provided for each wire at each junction box.

3.8.2 *Shipping cases.* Each cabinet shall have a platform and a case inclosing the unit and shall be constructed in accordance with Drawing 5–14–341. The case shall be fabricated of ½ inch thick plywood bonded to sheet of .0239 inch (24 gage) commercial quality cold rolled galvanized steel. The plywood shall be grade A–B, exterior type Douglas-fir plywood in accordance with commercial standard CS 45–60. The steel sheet shall be securely bonded to the grade B face with an adhesive which is noncorrosive to the metal face. The plywood shall be fastened around an inner frame of spruce or Douglas fir. The wood frame shall consist of upright corners firmly fastened together with longitudinal stringers to form a sturdy self-supporting framework. Over this framework, the plywood sheeting shall be applied to form a tight container, the steel side of the plywood being on the inside. All edges shall be covered with 2 inch by 2 inch .0478 inch (18 gage) angles with joints mitered and welded. Each case shall have six recessed handles securely bolted to the metal faced plywood casing and shall also have six recessed locking fasteners to fasten the casing to the platform. The platform shall be constructed of 2 inch nominal thickness tongue and grooved wood frame and firmly fastened to three 2 by 4 inch skids. Details of construction of case and platforms shall be as shown on the applicable drawing.

3.8.2.1 *Finish for shipping cases.* The angle frame of the shipping case shall be finished in the same manner as specified for the cabinet in 3.8.1.8. Wood surfaces of the shipping case shall be sanded smooth and then coated with primer conforming to 3.3.1. When dry, the primed surfaces shall be given a full hiding coat of enamel conforming to 3.3.2.

3.8.3 *Low temperature equipment.* The following low temperature refrigerated equipment shall be furnished with one set of operating instructions, maintenance instructions, parts list and wiring diagram permanently mounted in each unit and one set included in bookcase (unit No. 8).

Item No. 434 Viscosimeter, Kinematic
Item No. 437 Refrigerator
Item No. 438 Cloud and Pour Apparatus
Item No. 439 Channel Point Apparatus

3.9 **Mechanical equipment operation.** Each completely assembled cabinet containing mechanical equipment, such as refrigeration or air compressor equipment, electric motors and functioning assemblies shall operate smoothly, without undue vibration or noise. All controls shall operate and perform their designed functions when tested as specified in 4.3.1.

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3.10 Refrigeration equipment temperature. The refrigeration equipment shall have the capacity to lower the temperature within the cabinets to the specified degree indicated on the applicable Drawings and 4.3.1.

3.11 Marking for identification. Each unit shall be stamped or otherwise permanently marked in a plain and legible manner. The unit number shall be located in the front lower right hand corner. Shipping cases should be marked with the unit number and be marked in accordance with Standard MIL–STD–130.

3.12 Workmanship. The finished laboratory base shall conform to the quality and grade of product established by this specification. All pits, die-marks, burs and sharp edges shall be removed. There shall be no evidence of tacky finish, peeling, blistering, chipping or rust.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection. When a preproduction sample is required, it shall be examined for defects in 4.3.3.1, dimensions specified, and tested as specified in 4.4.1, 4.4.2, 4.4.3 and 4.4.4.

4.3 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be made in accordance with the provisions set forth in Standard MIL–STD–105, except where otherwise indicated hereinafter.

4.3.1 *Inspection of materials and components.* In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested and inspected in accordance with the specified requirements or referenced subsidiary specifications and standards to the extent specified herein or, if none, in accordance with this specification.

4.3.2 *Inspection at intermediate stages.*

4.3.2.1 *Intermediate inspection.* Processes, materials or assemblies that cannot be checked during inspection of the end item, i.e., installation of insulation in refrigerated cabinets, welding processes, annealing, painting, shall be examined or tests made at intermediate stages to determine compliance with applicable requirements. Any deviation from specified requirements shall be classified a defect. The inspection level shall be S–1 with an acceptable quality level of 4.0 defects expressed in terms of defects per hundred units.

4.3.3 *Inspection of the end product.*

4.3.3.1 *Visual examination.* Visual examination shall be made of each of the laboratory base cabinets including the laboratory equipment required to be furnished with each base unit. Examination shall be made for defects in design, material, construction, welding, workmanship, finish and marking including type, make, condition and securing of laboratory equipment within the respective base cabinets. Additional laboratory equipment, when specified in the contract or order, shall also be examined for compliance with applicable requirements including quantity and condition. Any evidence of noncompliance shall be cause for rejection of the cabinet or equipment as applicable. The following shall be used as a guide for the examination:

Examine	*Defect*
Cabinets: Finish of interior and exterior surfaces of cabinet and frame, including fume hood and frame.	Not painted where specified. Stainless steel cabinet tops and covers not polished, after welding and fabricated, to a No. 4 finish. Evidence of rust or corrosion. Enamel wrong type or color.

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Examine	Defect
	Dirt or other foreign matter imbedded in paint.
	Paint not adherent (blistered, lifted or peeled) or not completely dry, tacky, or oil under paint.
	Touchup not neatly accomplished.
	Surface not smooth and uniform.
	Thin film (primer shows through enamel).
	Paint scratch exposing primer or bare metal.
	Bolts, nuts, screws, any unpainted fasteners, not plated.
	Waste outlet on sink cabinet not chrome plated.
Cabinets: design construction and material	Details of design or construction not in accordance with applicable drawings, i.e., cabinets, fume hood, et cetera out of alignment (not square).
	Door hinge not continuous type, semiconcealed, fast pin (institutional type).
	Hinge not securely welded to the cabinet behind the upright corner posts.
	Channel clips not securely welded to back of door at point of fastening to the hinge leaf.
	Rabbetted recesses not provided for door closure.
	Rubber door bumpers missing from rabbetted recess.
	Framing members not welded (fastened together by means other than welding).
	Frame not reinforced around entire perimeter at bottom of cabinet with angle iron.
	Gusset plates (flanged on all sides) containing leveling bolts not provided at bottom of cabinet base and placed at the corners.
	Leveling bolt or bolts missing.
	Leveling bolts not accessible from interior of cabinet.
	Laboratory equipment, when in the respective storage place, not secure (subject to movement).
	Doors or drawer fronts not double wall pan type construction.
	Door heads not sound deadened.

Examine	Defect
	Drawer binds when moved in or out.
	Drawers not provided with four ball bearing glides.
	Doors and drawers not provided with flush type recessed drop handles with stainless spring loaded locking mechanism.
	Door or drawer will not lock securely in a closed position.
	Card index holder for drawer missing or not securely attached to drawer.
	Rubberized hair, rubber lining, forms and fastening devices not provided in drawers as applicable (see 3.7.1.5 and drawings).
	Underside of drawer trays not lined with sponge rubber.
	Rubber lined partitions between drawers not provided where specified (see drawings).
	Removable cover for cabinets do not fit properly.
	Service lines and supports (such as pipe, valves) missing or in wrong location (see 3.8.19) or wrong color code.
	Rack out of square, will not fit on cabinet.
	Elevating platforms not provided with ball bearing rollers.
	Platform binds or catches when raised or lowered.
	Platform locking mechanism does not lock securely in the raised or lowered position.
	Platform spring tension incorrect (load is unbalanced for the weight of equipment to be mounted on the platform).
	Elevated platform fastening devices do not provide firm and secure fastening of the equipment to the platform.
	Fastening devices on platform are such that the equipment is not easily removed for replacement of other equipment.
	Casters, with adjustable height screws, not provided for air compressor base.
	Interior of refrigerator cabinet not corrosion-resisting steel, seamless construction.
	Handles on refrigerator cabinet not recessed.
	Door gasket does not provide a

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Examine	Defect
	good seal (check with paper must have definite drag when withdrawn).
	Grille opening or motor compartment size not in accordance with drawing 5–14–337.
	Ice cube trays missing (12 trays)
	Less than 3 inches insulation space over motor compartment including thickness of metal sheets on refrigerated units.
	Sink (in cabinet sink) not stainless steel.
Fume hoods: Design construction and material.	Frame of hood not integral one piece welded construction.
	Steel panels not removable as indicated on applicable drawing.
	Sealed chamber in upper section of hood not provided to prevent fumes from entering blower compartment or not properly sealed, has open joints.
	Exhaust blower not mounted on a cork insulated platform or hard rubber pad.
	Transit or equal (asbestos board) cracked.
	Fan motor not size specified.
	Sash of hood not glazed with clear plexiglas or plexiglas not set in rubber.
	Cables and sash track not stainless steel.
	Pulleys not bronze.
	Hood out of alignment (not square).
	Blower housing not cast iron.
	Interior of fume hood not painted white.
Cabinets (general): Workmanship of bolts, nuts, screws, cotter pins and washers.	Bolt, nut, screw or washer missing where required.
	Bolt or screw head broken off.
	Bolt, nut or screw not drawn up tight where required.
	Bolt or screw in wrong location.
	Type or size of bolt, nut, screw, cotter pin or washer not as specified.
	Evidence that bolt, nut, screw or washer is not fabricated of material specified.
Workmanship of welds.	Weld (except spot weld) missing where required.
	Type of weld other than type specified.
	Spot weld missing where required.
	Weld completely broken or not fused.
	Weld not smooth and flush with adjacent material
	Spot welds (when applicable) not spaced as specified.
Workmanship on cabinets (general).	Component inoperative (when subject to manual movement).
	Burr or sharp edge that may cause injury.
	Ragged or chipped edge.
	Cabinet or equipment not clean.
Cabinets: Wiring, switches and panel board.	Switches broken or jammed.
	Switches loose.
	Loose wire connection.
	Wiring not inclosed in heavy rigid wall conduit.
	100 ampere circuit breaker not provided on panel board rated as specified, and with eight 1-pole, 30 ampere circuit breakers.
	Junction boxes, connection or wiring not in specified location.
	Positive identification of each wire not provided at each junction box.
	Wiring identification not the same in each unit.
	Insulation on wire cut or abraded.
	Connectors not secure.
	Wiring diagram missing from units where specified.
Basic components, equipment and accessories (Including additional laboratory equipment, when required).	Part damaged affecting serviceability.
	Equipment missing (except as otherwise classified herein).
	Equipment not in location specified.
	Component or equipment broken, fractured, or cracked.
	Part or equipment insecurely attached or not permanently attached where required.
	Equipment inoperative (when subject to manual movement).
	Component or equipment deformed, buckled, warped or bent out of shape.
	Equipment not size specified.
	Sharp edge or projection that may cause injury.
	Burr not removed.
	Ragged or chipped edges.
	Evidence that equipment is not of material specified.

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Examine	*Defect*
	Component or equipment not type or brand name specified (check for contracting officer's approval of other than brand name component or equipment).
	Component not formed as specified.
Identification marking of cabinets, name plates, parts and equipment list, et cetera	Missing, incomplete, illegible, not specified type or size, not in proper location, not affixed in prescribed manner, not correct color, not fabricated of specified material, not stamped, embossed, printed, lithographed, whichever is required.

4.3.3.2 *Dimensional examination.* Inspection shall be made of each cabinet for defects in dimensions. Any deviation exceeding the tolerance specified on drawings shall be cause for rejection.

4.3.3.3 *Testing of the end product.* Each completely assembled cabinet shall be tested in accordance with 4.4.1.

4.3.4 *Inspection of preservation, packaging, packing and marking.* Examination shall be made of preservation, packaging, packing and marking to determine compliance with specification requirements. The end item and shipping container shall be specifically checked for compliance with the following characteristics and tested as specified in 4.3.3:

(a) Units packed as specified in 3.8.2:
  (1) That all equipment is secure in their respective place so as not to be subject to breaking or damage.
  (2) Securing or locking of doors and damage.
  (3) Fastening of compressors with wood blocks so as not to ride on shock mounts and presence of instructions for removal of same.
  (4) Plywood is type specified (check plywood marking prior to painting).
  (5) Bonding of steel to plywood on B grade size (see 3.8.2).
  (6) Thickness of plywood and steel.
  (7) Securing of cabinets to base, size of bolts, (see Drawing 5–14–341).
  (8) Placement of recessed handles and fasteners.
  (9) Locking of fasteners to base and security of the lock.
  (10) Size of wood members, including base and skids.
  (11) Placement and size of bolts.
  (12) Presence of angle iron at outer corners of case and welding at mitered joints.
  (13) Presence of corrosion-resisting steel channel at bottom of container.
  (14) Painting of shipping container (interior and exterior) in accordance with 3.8.2.1 and that color is olive drab.
  (15) That shipping containers are relatively water and vapor-tight by sealing with waterproof tape (over all joints).
  (16) Fit of container cabinets.
  (17) Marking.

(b) Shipping case for additional equipment and supplies (see 5.1.1):
  (1) Size of members (see Drawing 5–14–348).
  (2) Armor ply conforms to Drawing 5–14–341 (see 4.3.4a, (5)) for vapor bath unit.
  (3) Presence of recessed carrying handles and locking devices, where required.
  (4) Securing or fastening of equipment or supplies.
  (5) That certain chemicals are not packed in the same container if specified (see note on Drawing 5–14–348).
  (6) Cushioning, bracing, blocking or bolting as applicable.
  (7) Presence of absorbent tablets in all boxes containing acid.
  (8) Painting of container, interior and exterior (vapor bath case).

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(9) Clearance between equipment and container is as specified when required.
(10) Skid type is provided when specified on drawing.
(11) That oversea type shipping containers are provided with waterproof case liners.
(12) Marking.

4.4 Tests.

4.4.1 *Operational test.* Each completely assembled cabinet containing mechanical equipment shall be operated to determine compliance with 3.9 and 3.10.

4.4.2 *Finish thickness tests.* One of each type cabinet shall be tested to determine compliance with the total film thickness requirement specified in 3.8.1.8. Three determinations shall be made for each surface (exterior and interior surfaces). Any determination not meeting the required total film thickness shall be cause for rejection of the cabinets represented by the defective cabinet.

4.4.3 *Testing of doors.* Each cabinet door shall be opened and closed approximately 10 times with a 150 pound weight suspended from the top edge of door. Failure of door to conform with the requirement specified in 3.8.1.4 shall be cause for rejection.

4.4.4 *Testing of packed units.* Prior to applying tape to seams of the shipping containers, one completely packed cabinet of each type shall be tilted 35° and released to free fall on the skid base. This shall be performed from each of the four sides, after which the shipping case shall be removed from its base and the cabinet and contents examined to determine that fastening devices and packing materials are adequate to prevent breakage or displacement of apparatus (laboratory equipment). Evidence of unsatisfactory fastening devices or packing material shall be corrected on all like units failing the test.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A or C as specified (see 6.2).

5.1.1 *Level A.* No packing shall be required except that all joints of the shipping container and supporting platform shall be taped with 2-inch minimum width tape conforming to type I or II, class 1 of Specification PPP–T–60.

5.1.1.1 *Additional equipment and supplies.* When additional equipment and supplies are required for which reinforced plywood containers have not been constructed, they shall be packed in wood containers as shown on Drawing 5–14–348 with a type I or II, grade C case liner conforming to Specification MIL–L–10547. Containers shall be painted olive drab conforming to Specification TT–E–529.

5.1.2 *Level C.* Units shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.2 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Standard MIL–STD–129. Set markings as stipulated in Standard MIL–STD–129, shall be used.

6. NOTES

6.1 Intended use. The base laboratory is used for the performance of quality surveillance of all petroleum products.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title and date of this specification.
(b) Whether preproduction sample is waived (see 3.2).
(c) When C.O.T. Military Engine is required (see 3.4).
(d) Whether or not chemicals specified in table II are to be furnished (see 3.5, 3.6 and 6.3).
(e) When table V items are required

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for other than Air Force procurement (see 3.5).

(f) Selection of applicable level of packing (see 5.1).

6.3 Because chemicals deteriorate in storage, the requisitioner should notify the Purchasing and Contracting officer when chemicals listed in table II are required.

6.4 This specification, together with the indicated drawings and the tables showing a list of laboratory equipment and supplies, form the basis for the complete assembly required. There are approximately 84 separate packages in each assembly, three of which do not require a specially constructed shipping case. These three are the packages making up the knock testing unit which are shipped in the cases supplied by the manufacturer of this unit. Gas cylinders and 55 gallon drums do not require overpacking or other than commercial painting.

Custodians:
Army—GL
Air Force—MAA
Interested activities:
Review
Army—MD, MU

Preparing activity:
Army—GL
Project No. 6640–0236

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TABLE I. *Equipment for One Complete Base Petroleum Laboratory*
(Trade Names and Catalog Numbers are for Reference Only)
(or equal) After Each Reference

No.	Nomenclature	Unit	Amount required
1	Aniline Point Apparatus, S.I.L. U-Tube Model, Complete Ref: Precision Scientific Co. No. 73478	set	1
2	U-Tube, Glass, S.I.L. Model Ref: Precision Scientific Co. No. 330012	ea	3
3	Thermometer, ASTM, Range 25–105 C., Aniline Point Apparatus Ref: Emil Greiner Co. No. 23534	ea	6
4	Heater, Ful-Kontrol, 750 Watts, 115 V, 50–60 Cycles, A.C. Ref: Precision Scientific Co. No. 73482	ea	1
5	Conradson Carbon Residue Apparatus, Complete Ref: Precision Scientific Co. No. 73571	set	2
6	Outer-Crucible and Cover, Nickel, Carbon Residue Apparatus Ref: Precision Scientific Co. No. 68176	ea	1
7	Crucible, Porcelain, Coor Carbon Res. Apparatus Ref: Type I, Size 4, NNN–C–680	ea	12
8	Crucible, Skidmore, w/Nickel Cover, Carbon Res. Appar Ref: Precision Scientific Co. No. 68245	ea	1
9	Hood, Nickel, Carbon Residue Apparatus Ref: Precision Scientific Co. No. 73585	ea	1
10	Triangle, Chromel, (a) Carbon Residue Apparatus Ref: NNN–T–685, Type II	ea	8
11	Burner, Gas Laboratory for Carbon Residue Apparatus, High Temperature Brass Ref: Spec NNN–B–795 Type I *Note.* Item numbers underlined indicate complete apparatus. Subsequent numbers not underlined indicate spare parts or general supplies for the underlined item above.	ea	1
12	Beads, Glass, 5 mm. dia. Carbon Residue Apparatus Ref: Spec MIL–B–3473	lbs	1
13	Block, Refractory, Molded, Porcelain Gar. Res. Appar Ref: Precision Scientific Co. No. 73577	ea	1
14	Centrifuge, Electric, Complete Ref: International Model H. No. 470	set	1
15	Cups, Aluminum, Pear Shape, 100 ml. Centrifuge Ref: E. Greiner No. G 4729M	ea	4
**16	Head Ref: International No. 204	ea	1
17	Tube, Centrifuge, Grad. 100 ml Ref: E. Greiner No. G 4945	ea	12
18	Chromometer, Saybolt, Complete Ref: Emil Greiner No. G 17585	set	1
19	Tube, Glass, Plain, Chromometer, Saybolt Ref: Emil Greiner No. G 17595	ea	2
20	Tube, Glass, Graduated, Chromometer, w/Connection Ref: Emil Greiner No. G 17592	ea	2
21	Gasket, Drain, Spout, Chromometer Ref: Emil Greiner No. G 17585	ea	12
22	Thermometer, Range —36 F to +120 F Cloud and Pour Point Apparatus Ref: E. Greiner G 23546	ea	6
23	Jar, Pour Test (2 Calorimeter) Ref: Precision Scientific No. 74530	ea	12

For explanation of footnotes see end of table.

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TABLE I. *Equipment for One Complete Base Petroleum Laboratory*
(Trade Names and Catalog Numbers are for Reference Only)
(or equal) After Each Reference—Continued

No.	Nomenclature	Unit	Amount required
24	Cloud and Pour Point Apparatus, 4 Unit, All Metal Ref: Emil Greiner No. G 17485	set	1
25	Thermometer, Range —112 F to +70 F, Cloud and Pour Point Apparatus Ref: Fisher Scientific Co. No. 13–412	ea	12
26	Colorimeter, Tag-Robinson, Complete Ref: Emil Greiner No. G 17625	set	1
27	Colorimeter, ASTM D 1500 Ref: Emil Greiner No. G 17641	set	1
28	Distillation Apparatus, Front View, w/750 Watt Lo-Cap Heater 115V, A.C., Gasoline, Comm. Set Ref: Emil Greiner No. G 17880	set	3
29	Board, Asbestos, 4" Hole, 8 x 11 x ¼" Ref: Emil Greiner No. G 17825L	ea	2
30	Board, Asbestos, 2¾" Hole, 6" x 6" x ¼" Ref: Emil Greiner No. G 17825 E	ea	3
31	Board, Asbestos, 1½" Hole, 6" x 6" x ¼" Ref: Emil Greiner No. G 17825D	ea	4
32	Board, Asbestos, 1¼" Hole, 6" x 6" x ¼" Ref: Emil Greiner No. G 17825	ea	4
33	Cylinder, Single Grad., Glass 200 ml 0–100% Special Ref: Emil Greiner No. G 6921	ea	12
34	Jar, Battery, 4⅝" Dia x 9" High for Dist. Appar Ref: Emil Greiner No. G 14635	ea	1
35	Thermometer, Range 30 F–580 F., Distill. Apparatus Ref: Emil Greiner No. G 23552	ea	12
36	Thermometer, Range 30 F–760 F., Distill. Apparatus Ref: Emil Greiner No. G 23556	ea	6
37	Washer, Rubber, Distill. Appar. 1½" square 1/16" thick with ½" hole in center Ref: Palo Laboratory Supplies No. T8630–NR	ea	12
38	Flask, Distilling, 100 cc., Distillation Appar Ref: NNN–F–240, Type III, class 1, Style II, Size 1	ea	24
39	Flask, Distilling, 250 cc., Distillation Appar Ref: NNN–F–240, Type III, class 1, Style II, Size 2	ea	12
40	Dropping Point Apparatus, Complete w/550 Watt Heater Ref: Precision Scientific No. 73455	set	1
41	Grease Cup Assembly, Dropping Point Apparatus Ref: Precision Scientific No. 73457	ea	3
42	Test Tube for Dropping Point Apparatus Ref: Precision Scientific No. 73461	ea	6
43	Thermometer, Range 20–580 F., Dropping Point Ref: Precision Scientific No. 72820F	ea	6
44	Flash and Fire Point, Cleveland Open Cup, Precision Ful-Kontrol Appar., Comp. w/750 Watt Heater—115V. 50–60 Cycles, A.C. Ref: Precision Scientific No. 74582	set	1
45	Thermometer, Range 20–760 F., Flash Point, COC Ref: E. Greiner No. G 23572	ea	6
46	Flash Point, Pensky-Martens, Closed Cup, Complete Ref: Emil Greiner No. G 18196	set	1
47	Thermometer, ASTM, Range 20–230 F., Pensky-Martens Ref: Emil Greiner No. G 23576	ea	6

For explanation of footnotes see end of table.

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TABLE I. *Equipment for One Complete Base Petroleum Laboratory*
(Trade Names and Catalog Numbers are for Reference Only)
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No.	Nomenclature	Unit	Amount required
48	Thermometer, ASTM, Range 200–700 F., Pensky-Martens Ref: E. Greiner No. G 23580	ea	4
49	Flash Point, Tag, Closed Tester, ASTM, w/Alcohol Burner, Complete Ref: Emil Greiner No. G 18180	set	1
50	Thermometer, Closed Tester, Range —4 to 122 F. Ref: Emil Greiner No. 23588	ea	12
51	Gum content of Gasoline Appar. ASTM, 110V., Complete Ref: Precision Scientific No. 74775	set	1
52	Beaker, Laboratory; Glass 100ml Capacity Ref: Type III, Spec NNN–B–175	ea	24
**53	Oxidation Stability Apparatus, Gasoline (2 hole bath) 115 Volt, A.C. Ref: (E. Greiner No. G 18550 Bath) Complete with the following:	set	1
	(1) Bomb, E. Greiner No. G 18500	ea	2
	(2) Wrench, E. Greiner No. G 18502	ea	1
	(3) Table Socket, E. Greiner No. G 18503	ea	1
	(4) Recorded, 2-Pen w/charts, E. Greiner No. 18525	ea	1
	(5) Ink 2 oz. Bottle, Red, E. Greiner No. 19625	ea	1
	(6) Ink 2 oz. Bottle, Blue, E. Greiner No. 19625/1	ea	1
	(7) Flexible Metal Tubing (Bomb to Recorder) E. Greiner No. 18545	ea	2
	(8) Liner, Glass w/Cap, E. Greiner No. G 18512	ea	2
54	Clamp, Bench, Gum Stability Apparatus Ref: Emil Greiner No. G 18503	ea	1
55	Liner, Glass, Gum Stability Apparatus Ref: Emil Greiner No. G 18510	ea	24
56	Recording Pressure Gage, 12", Range 0–200 lbs in 2 lb. divisions, 2 pen type. Ref: Emil Greiner No. 18525	ea	2
57	Hose, Rubber, Air w/Air Check Valve, 6 ft Lengths for Gum Stability-Apparatus Red-Patch and Reliner Com. Ref: Schrader No. 5495 (Brooklyn, N.Y. Paramount Sales Co.)	ea	1
58	Thermometer, Range 204–218 F., Gum Stab Apparatus Ref: Fisher No. 13–575 (22F)	ea	6
59	Washer, Neoprene, Gum Stability Apparatus Ref: Emil Greiner No. G 18515	ea	24
60	Bomb, Complete, Gum Stability Apparatus Ref: Emil Greiner No. G 18500	ea	2
61	Charts, Pressure Gage, Gum Stability Apparatus Ref: Emil Greiner No. G 18525	box	2
62	Ink Recording, 2 Oz. bottle (3 ea Red and 3 ea Blue) Gum Stab. Appar Ref: Emil Greiner Co. No. G 19625 Red and G 19625/1 Blue	btl	6
63	Tubing, Flexible, Metal (Bomb and Recorded Connection) Ref: Emil Greiner No. G 18545	ea	4
64	Wrench, Bomb, Gum Stability Apparatus Ref: Emil Greiner No. G 18502	ea	2
65	Sulfur Bomb, Petroleum Oils, Parr. Elec. Ignition ASTM–D–129–39, Complete Ref: Central Scientific No. 27808	set	1

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TABLE I. *Equipment for One Complete Base Petroleum Laboratory*
(Trade Names and Catalog Numbers are for Reference Only)
(or equal) After Each Reference—Continued

No.	Nomenclature	Unit	Amount required
66	Thermo-Hydrometer, Large, 0–11 degrees Ref: Emil Greiner No. 12650	ea	2
67	Thermo-Hydrometer, Large, 10–21 degrees Ref: Emil Greiner No. 12652	ea	3
68	Thermo-Hydrometer, Large, 19–31 degrees Ref: Emil Greiner No. 12654	ea	4
69	Thermo-Hydrometer, Large, 29-41 degrees Ref: Emil Greiner No. 12656	ea	4
70	Thermo-Hydrometer, Large, 39–51 degrees Ref: Emil Greiner No. 12658	ea	4
71	Thermo-Hydrometer, Large, 49–61 degrees Ref: Emil Greiner No. 12660	ea	8
72	Thermo-Hydrometer, Large, 59–71 degrees Ref: Emil Greiner No. 12662	ea	4
73	Thermo-Hydrometer, Large, 69–81 degrees Ref: Emil Greiner No. 12664	ea	2
74	Thermo-Hydrometer, Large, 79–91 degrees Ref: Emil Greiner No. 12666	ea	2
75	Thermo-Hydrometer, 4 oz. size 5–11 degrees Ref: E. Greiner 48–552	ea	2
76	Thermo-Hydrometer, 4 oz. size 10–16 degrees Ref: E. Greiner 48-554	ea	2
77	Thermo-Hydrometer, 4 oz. size 15–21 degrees Ref: E. Greiner 48–556	ea	4
78	Thermo-Hydrometer, 4 oz. size 20–26 degrees Ref: E. Greiner 48–558	ea	4
79	Thermo-Hydrometer, 4 oz. size 25–31 degrees Ref: E. Greiner 48–560	ea	4
80	Thermo-Hydrometer, 4 oz. size 30–36 degrees Ref:E. Greiner 48–562	ea	4
81	Thermo-Hydrometer, 4 oz. size 35–41 degrees Ref: E. Greiner 48–564	ea	4
82	Thermo-Hydrometer, 4 oz. size 40–46 degrees Ref: E. Greiner 48–566	ea	4
83	Thermo-Hydrometer, 4 oz. size 45–51 degrees Ref: E. Greiner 48–568	ea	4
84	Thermo-Hydrometer, 4 oz. size 50–56 degrees Ref: E. Greiner 48–570	ea	4
85	Thermo-Hydrometer, 4 oz. size 55–61 degrees Ref: E. Greiner 48–572	ea	8
86	Thermo-Hydrometer, 4 oz. size 60–66 degrees Ref: E. Greiner 48–574	ea	4
87	Thermo-Hydrometer, 4 oz. size 65–71 degrees Ref: E. Greiner 48–576	ea	2
88	Thermo-Hydrometer, 4 oz. size 70–76 degrees Ref: E. Greiner 48–578	ea	2
89	Penetrometer, Complete Ref: Precision Scientific No. 73510	set	1
90	Cone, Grease Ref: Precision Scientific No. 73526	ea	4
91	Grease Worker Ref: Precision Scientific No. 73531	ea	1
92	Needle, Penetrometer Ref: Precision Scientific No. 73520	ea	2

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TABLE I. *Equipment for One Complete Base Petroleum Laboratory (Trade Names and Catalog Numbers are for Reference Only) (or equal) After Each Reference*—Continued

No.	Nomenclature	Unit	Amount required
93	Sediment by Extraction Apparatus, Complete	set	2
—	Ref: Emil Greiner No. G 19145		
94	Condenser, Metal, Extraction	ea	2
	Ref: E. Greiner No. 6480		
95	Flask, Pyrex, Sediment by Extraction	ea	6
	Ref: Type IV, class 1, Style 2, Size 3, Spec NNN–F–240		
96	Support, Thimble, Sediment by Extraction	ea	6
	Ref: E. Greiner No. G 19160		
97	Thimble, Alundum, Sediment by Extraction	ea	12
	Ref: Emil Greiner No. G 19150		
98	Steam Emulsion Apparatus, Complete	set	2
—	Ref: Fisher 13–488		
99	Tubes, Oil, Emulsion Pyrex	ea	6
	Ref: Fisher 13–488G		
100	Jar, Pyrex, Steam Emulsion	ea	4
	Ref: Fisher 13–488D		
101	Thermometer, Range 0–230 F, Steam Emulsion	ea	4
	Ref: Fisher 13–990B		
102	Thermometer, Range 30–220 F, Steam Emulsion	ea	4
	Ref: Fisher 13–488–E		
103	Smoke Point Apparatus, Complete	ea	1
—	Ref: E. Greiner No. G17390		
104	Wicks for Smoke Point Apparatus	gross	1
—	Ref: E. Greiner No. G17392		
105	Sulfur Determination Appar. (Lamp Method) Complete	ea	2
	Ref: Emil Greiner G19320 with G19332		
106	Absorber, Pyrex, Sulfur, Determination Apparatus	ea	2
	Ref: Emil Greiner G19322		
107	Burner for Sulfur Determination Apparatus	ea	2
	Ref: Emil Greiner G19328		
108	Chimney, Pyrex, Sulfur Determination Apparatus	ea	2
	Ref: Emil Greiner G19326		
109	Flask, Lamp, Pyrex, Sulfur Determin. Apparatus	ea	5
	Ref: Emil Greiner G19324		
110	Trap, Spray, Ground Joint	ea	2
	Ref: Emil Greiner G19330		
111	Wicks for Sulfur Determination Apparatus	gross	1
	Ref: Emil Greiner G19334		
112	Stand, Sulfur Determination Apparatus	ea	2
	Ref: Emil Greiner G19332		
113	Tetraethyl Lead Extraction Apparatus (Stand and Rods)	set	3
—	Ref: Emil Greiner G19390		
114	Viscosimeter, Thermo, Saybolt	ea	1
—	Ref: Emil Greiner G20060		
115	Vapor Pressure Apparatus, Complete, less bombs and thermometer	set	1
—	Ref: Emil Greiner G19530		
**116	Bomb, Vapor Pressure, ¼ size	ea	3
	Ref: Greiner No. G19485		
**117	Gage, Vapor Pressure, (for ¼ size bomb) 2½" 0–15 lbs	ea	3
	Ref: E. Greiner No. G19486		
118	Bomb, Vapor Pressure	ea	3
	Ref: Emil Greiner G19476		

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TABLE I. *Equipment for One Complete Base Petroleum Laboratory (Trade Names and Catalog Numbers are for Reference Only) (or equal) After Each Reference*—Continued

No.	Nomenclature	Unit	Amount required
119	Gage, Vapor Pressure, Range 0–15 lbs. Ref: Emil Greiner G19505	ea	4
120	Gage, Vapor Pressure, Range 0–45 lbs. Ref: Emil Greiner G19510	ea	1
121	Thermometer, Range 94–108 F Ref: Emil Greiner G23640	ea	6
122	Thermometer, Range —30 to 120 F Ref: Emil Greiner G23608	ea	6
123	Viscosimeter, Kinematic, Complete Ref: E. Greiner No. G19600	set	2
124	Pan, w/Lip, 1 Qt. Capacity, Viscosimeter Ref: Fisher 13–584	ea	1
125	Strainer, Oil, Viscosity Ref: Fisher 13–583	ea	2
126	Standardization Oil, 100 F, 122 F and 210 F Ref: ASTM Viscosity Oil Standard S–600	qts	4
127	Thermometer, Range 97.5–102.5 F., Viscosity Kinematic Ref: E. Greiner No. G23626	ea	12
128	Thermometer, Range 127.5–132.5 F., Viscosity Kinematic Ref: E. Greiner Co. No. G23630	ea	12
129	Thermometer, Range 207.5–212.5 F., Viscosity Kinematic Ref: E. Greiner No. G23636	ea	12
130	Pipette, Kinematic, Fenske-Ostwald, ASTM, Calibrated, No. 50 Ref: E. Greiner No. G19620M	ea	2
131	Pipette, Kinematic, Fenske-Ostwald, ASTM, Calibrated, No. 100 Ref: E. Greiner No. G19620N	ea	2
132	Pipette, Kinematic, Fenske-Ostwald, ASTM, Calibrated, No. 200 Ref: E. Greiner No. G19620Q	ea	2
133	Pipette, Kinematic, Fenske-Ostwald, ASTM, Calibrated, No. 300 Ref: E. Greiner No. G19620R	ea	2
134	Pipette, Kinematic, Fenske-Ostwald, ASTM, Calibrated, No. 400 Ref: E. Greiner No. G19620T	ea	2
135	Pipette, Kinematic, Fenske-Ostwald, ASTM, Calibrated, No. 500 Ref: E. Greiner No. G19620V	ea	2
136	Pipette, Viscometer, Cannon-Fenske, Opaque, No. 600 CO, Calibrated Ref: E. Greiner No. G19622Z	ea	4
137	Thermometers, Double Diamond Brand, Engraved Stem, Mercury Filled, Range —30 to 0 F., Calibrated Ref: H. B. Instrument Co., Cat. No. CSPRF–15	ea	6
138	Thermometers, Double Diamond Brand, Engraved, Stem, Yellow Back, Mercury Thallium Filled, Range —61 to —29 F., Cal. Ref: H. B. Instrument Co., Cat. No. 21339–C	ea	6
139	Thermometers, Double Diamond Brand, Engraved Stem, Red Liquid Filled, Range —90 to 60 F., Calibrated Ref: H. B. Instrument Co., Cat. CSK 23	ea	6

For explanation of footnotes see end of table.

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TABLE I. *Equipment for One Complete Base Petroleum Laboratory (Trade Names and Catalog Numbers are for Reference Only) (or equal) After Each Reference*—Continued

No.	Nomenclature	Unit	Amount required
140	Rod, Glass, Borosilicate, 6 mm. 4 ft length Ref: Spec. NNN–R–560, Type I	lbs	5
141	Rod, Glass, Borosilicate, 8 mm. 4 ft length Ref: Spec. NNN–R–560, Type I	lbs	5
142	Rod, Glass, Borosilicate, 4 mm. 4 ft length Ref: Spec. NNN–R–560, Type I	lbs	5
143	Rod, Glass, Borosilicate, 10 mm. 4 ft length Ref: Spec. NNN–R–560, Type I	lbs	5
144	Wire, Gauze, 16 Gage Asbestos, 6 x 6 Ref: Spec. NNN–G–698, Type II, Grade D, Size 3	ea	6
145	Water Determination Appar., Complete w/Elec. Heater, 115 V, 550 Watts Ref: E. Greiner No. G 20171	set	2
146	Condenser, Pyrex, 400 mm Ref: Fisher 7–720	ea	4
147	Receiving Trap, Water by Distillation Ref: E. Greiner No. G7525	ea	6
148	Flask, Pyrex, 500 ml. Round Bottom, Long Neck Ref: Spec. NNN–F–240, Type I, Style I	ea	6
149	Distilling Receiver, Crank Case Dilution Ref: E. Greiner No. G7507	ea	6
150	Asbestos, Fibre Acid, Washed, Short Fibre Ref. MIL–F–50026	lbs	1
151	Balance, Analytical, Voland #640 Ref: E. Greiner No. G1090	ea	1
152	Weight, Balance 1 mg. –100 gr. Analytical, Complete Ref: Greiner No. G1446	set	2
153	Balance, Analytical, Voland #220 Ref: E. Greiner No. G1072	ea	1
154	Balance, Torsion Ref: Fisher 2–080	ea	1
155	Weights, Balance 1 gr–500 gr Ref: Fisher 2–300	set	1
156	Barometer, Aneroid Ref: Fisher 2–405	ea	1
157	Bath, Water, Thermostatically Controlled, Complete Ref: Precision Scientific No. 66648	ea	1
158	Beaker, Griffin, 20 ml. Cap Ref: Specification NNN–B–175, Type I	ea	12
159	Beaker, Griffin, 50 ml. Cap Ref: Specification NNN–B–175, Type I	ea	12
160	Beaker, Griffin, 100 ml. Cap Ref: Specification NNN–B–175, Type I	ea	12
161	Beaker, Griffin, 250 ml. Cap Ref: Specification NNN–B–175, Type I	ea	12
162	Beaker, Griffin, 400 ml. Cap Ref: Specification NNN–B–175, Type I	ea	24
163	Beaker, Griffin, 600 ml. Cap Ref: Specification NNN–B–175, Type I	ea	24
164	Beaker, Griffin, 1000 ml. Cap Ref: Specification NNN–B–175, Type I	ea	12
165	Beaker, Griffin, 30 ml. Cap Ref: Specification NNN–B–175, Type I	ea	12

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TABLE I. *Equipment for One Complete Base Petroleum Laboratory*
(Trade Names and Catalog Numbers are for Reference Only)
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No.	Nomenclature	Unit	Amount required
166	Beaker, Griffin, 5 ml. Cap Ref: Specification NNN–B–175, Type I	ea	12
167	Beaker, Griffin, 800 ml. Cap Ref: Specification NNN–B–175, Type I	ea	12
168	Borer, Cork Ref: Specification NNN–B–600, Grade B, 15 per set	set	2
169	Bottle, Clear Glass, Screw Cap, Narrow Mouth, French Square, 8 oz. Ref: Fisher 2–882	ea	24
170	Bottle, Narrow Mouth, Clear Bakelite, Screw Cap, French Square, 1 Qt. Cap Ref: Fisher 2–882	ea	24
171	Bottle, Narrow Mouth, Clear, Round 1 Qt. Cap Cork Top Ref: Fisher 2–880	ea	48
172	Bottle, Narrow Mouth, Bakelite, Screw Cap, Amber, French Square, 32 oz. Ref: A.H.T. 2207–K	ea	48
173	Bottle Babcock, Unsaturation Test, Stoddard Solvent Ref: Fisher 13–559	ea	6
174	Bottle, Wide Mouth, Clear, Glass, Screw Cap, 4 oz. Ref: A.H.T. 6225–A	ea	48
175	Bottle, Washing, 500 ml. cap w/top Ref: MS 36069–1	ea	6
176	Bottle, Clear, Screw Cap, Round, Narrow Mouth, 2 oz Ref: Fisher 2–880	ea	12
177	Bottle for Oil Sampling, Clear, Long, 4 oz Ref: Fisher 3–086	ea	24
178	Bottle, Narrow Mouth, 5 gal. cap., Clear, Round Ref: Cenco 10310	ea	4
179	Bottle Reagent, Narrow Mouth, Clear Glass, 2 oz Ref: A.H.T. 2207–K	ea	12
180	Bottle, Reagent, Drop, CP, Narrow Mouth, Brown, Hood Stopper 60 ml. Ref: E. Greiner No. G 2480	ea	36
181	Bottle, Reagent, w/Glass Stoppers, Clear Glass, 16 oz. Round Ref: (Greiner 2275)	ea	36
182	Brushes, Flask and Bottle Ref: Fisher 3–571	ea	6
183	Brushes, Beaker, Large Ref: Specification H–B–173, Type I	ea	6
184	Brushes, Beaker, Small Ref: Fisher 3–565	ea	3
185	Brushes, Large, Test Tube Ref: Specification H–B–173, Type V, Style A, Size 1¼	ea	6
186	Brushes, Small, Test Tube Ref: Specification H–B–173, Type V, Style A, Size ¾	ea	6
187	Brushes, Burette, Small Ref: Fisher 3–614	ea	6
188	Brushes, Pipette Ref: Specification H–B–173, Type IV	ea	6

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TABLE I. *Equipment for One Complete Base Petroleum Laboratory*
(Trade Names and Catalog Numbers are for Reference Only)
(or equal) After Each Reference—Continued

No.	Nomenclature	Unit	Amount required
189	Brushes, Balance Ref: Fisher 3-654	ea	6
190	Brushes, Sink Ref: Fisher 3-682	ea	2
191	Brushes, Wire Ref: Fisher 3-685	ea	2
192	Brushes, Cylinder, Small Ref: Specification H-B-173, Type V, Style A, Size 1¾	ea	2
193	Brushes, Cylinder, Large Ref: Specification H-B-173, Type V, Style A, Size 2¼	ea	6
194	Bulb, Red, Rubber Ref: MS 36074-1	ea	12
195	Burette, 50 ml. cap. w/Stopcock, Blue Line "EXAX" Retested Ref: Specification DD-V-581, Type II, Class A, Grade 2	ea	12
196	Burner, Wing, Top, 11 mm. O. D. Ref: Fisher 3-995	ea	3
197	Burner, Gas Ref: Specification NNN-B-795, Type I, Artificial Gas (150-600 B.t.u.)	ea	6
198	Burner, High Temperature Ref: Specification NNN-B-795, Type I (2400-3300 B.t.u)	ea	12
199	Burner, Pittsburgh, Universal Ref: Type II, Liquefied petroleum NNN-B-795	ea	6
200	Burner, Alcohol Ref: Fisher 4-245	ea	2
201	Burner, Micro Ref: Fisher 20-141	ea	2
202	Burner, Forbes, Bunsen Ref: Type II, Artificial Gas NNN-B-795	ea	6
203	Can, Screw Top, 1 gal. Cap Ref: Fisher 4-667	ea	48
204	Can, Screw Top, 5 gal. Cap Ref: Fisher 4-667	ea	6
205	Can, Friction Top, 2 lb Cap (for Grease) Ref: American Can Company	ea	12
206	Can, Tin, 1 Qt. Cap. w/Screw Neck and Cap w/Innerseal (for lubricating oils) Ref: American Can Co.	ea	160
207	Cement, Dekhotinsky, 1 oz. pkg. Medium Ref: Fisher 4-749	pkg	6
208	Cement, Dekhotinsky, 1 oz. pkg. Hard Ref: Fisher 4-749	pkg	6
209	Cement, Duco, 1 oz. tubes Ref: Fisher 4-752	tubes	6
210	Charts, Viscosity Temperature, Viscosity Index Ref: E. Greiner No. G-20147	pads	2
211	Chamois, 14 x 18 inches (for filtering gasoline) Size B Ref: Fisher 5-680	ea	3
212	Charts, Kinematic, Viscosity, Type "E" Pads Ref: Emil Greiner No. G-20132	pads	2
213	Clamp, Beaker, Rubber Lined Ref: Cenco 12090	ea	6

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TABLE I. *Equipment for One Complete Base Petroleum Laboratory (Trade Names and Catalog Numbers are for Reference Only) (or equal) After Each Reference*—Continued

No.	Nomenclature	Unit	Amount required
214	Clamp, Double, Burette, Holder Ref: Specification GG-C-411, Type 1, Composition I	ea	4
215	Holder, Clamp, Castaloy Ref: Specification GG-C-411, Type 7, Composition I	ea	12
216	Clamp, Extension, Castaloy Ref: Specification GG-C-411, Type 4, Composition I	ea	12
217	Clamp, Versatile, Castaloy Ref: Specification GG-C-411, Type 5, Composition I	ea	6
218	Clamp, Burette, Adjustable, Castaloy Ref: Specification GG-C-411, Type 2, Composition I	ea	12
219	Clamp, Castaloy, Hosecock Ref: Specification GG-C-418, Type III, Class A, Size 2	ea	24
220	Clamp, Pinchcock, Brass, Flat Jaws, 2¾" Ref: Specification GG-C-418, Type I, Class A, Size 1	ea	12
221	Cloth, Cheese Ref: Specification CCC-C-440, Type II, Class 2	yds	100
222	Timer, Victor, Interval, Manual Ref: Emil Greiner No. G25050	ea	1
223	Cloth, Emery, No. 00 Ref: Fisher 9-535	sheet	24
224	Cotton, Absorbent, 12" Wide (1 lb. pkg.) Ref: Specification JJJ-C-561, Grade A, Class II, Size 4	pkg	3
225	Stopper, Cork, Regular Length, No. 2 Ref: Specification LLL-S-731, Type I, Grade A, Class 2	bags	1
226	Stopper, Corks, Regular Length, No. 3 Ref: Specification LLL-S-731, Type I, Grade A, Class 2	bags	1
227	Stopper, Corks, Regular Length, No. 4 Ref: Specification LLL-S-731, Type I, Grade A, Class 2	bags	1
228	Stopper Corks, Regular Length, No. 5 Ref: Specification LLL-S-731, Type I, Grade A, Class 2	bags	1
229	Stopper, Corks, Regular Length, No. 6 Ref: Specification LLL-S-731, Type I, Grade A, Class 2	bags	1
230	Stopper, Corks, Regular Length, No. 7 Ref: Specification LLL-S-731, Type I, Grade A, Class 2	bags	1
231	Stopper, Corks, Regular Length, No. 8 Ref: Specification LLL-S-731, Type I, Grade A, Class 2	bags	1
232	Stopper, Corks, Regular Length, No. 9 Ref: Specification LLL-S-731, Type I, Grade A, Class 2	bags	1
233	Stopper, Corks, Regular Length, No. 10 Ref: Specification LLL-S-731, Type I, Grade A, Class 2	bags	1
234	Stopper, Corks, Regular Length, No. 11 Ref: Specification LLL-S-731, Type I, Grade A, Class 2	bags	1
235	Stopper, Corks, Regular Length, No. 12 Ref: Specification LLL-S-731, Type I, Grade A, Class 2	bags	1
236	Stopper Corks, Regular Length, No. 14 Ref: Specification LLL-S-731, Type I, Grade A, Class 2	bags	1
237	Stopper, Corks, Regular Length, No. 15 Ref: Specification LLL-S-731, Type I, Grade A, Class 2	bags	1
238	Stopper, Corks, Regular Length, No. 18 Ref: Specification LLL-S-731, Type I, Grade A, Class 2	bags	1
239	Stopper Corks, Regular Length, No. 20 Ref: Specification LLL-S-731, Type I, Grade A, Class 2	bags	1
240	Stopper Corks, Regular Length, No. 22 Ref: Specification LLL-S-731, Type I, Grade A, Class 2	bags	1

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TABLE I. *Equipment for One Complete Base Petroleum Laboratory*
(Trade Names and Catalog Numbers are for Reference Only)
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No.	Nomenclature	Unit	Amount required
241	Stopper Corks, Regular Length, No. 24 Ref: Specification LLL–S–731, Type I, Grade A, Class 2	bags	1
242	Stopper Corks, Regular Length, No. 26 Ref: Specification LLL–S–731, Type I, Grade A, Class 2	bags	1
243	Sharpener, Cork, Borer Ref: Fisher 7–865	ea	1
244	Cork, Press, Rotary Ref: Specification GGG–C–589	ea	1
245	Cork, Screw Ref: Specification GGG–C–591, Type III	ea	2
246	Cord, Asbestos, Diameter 1/16"—lb Ref: Fisher 1–455	lbs	2
247	Cots, Finger, Asbestos, Style A for Fingers Ref: Fisher 9–998	ea	4
248	Cots, Finger, Style B for Thumbs Ref: Fisher 9–998	ea	2
249	Crucibles, Coors, 30 ml. Ref: Specification NNN–C–680, Type I, Size 4	ea	12
250	Crucibles, Coors, 50 ml. Ref: Specification NNN–C–680, Type I, Size 5	ea	12
251	Crucibles, Coors, Size #4, 150 ml. Ref: Fisher 7–955	ea	6
252	Crucible, Coors, Gooch, Porcelain, 25 ml. Ref: Specification NNN–C–680, Type 2	ea	12
253	Crucible, Holder, for size #3, Gooch Crucible Ref: Specification MIL–C–20033	ea	6
254	Crucible, Porcelain, Coors, 30 ml. High Form, Size #1 Ref: Fisher 7–965	ea	12
255	Cutter, Glass, Griffin Type Ref: Fisher 11–340	ea	6
256	Cylinder, Liquefied Petroleum, Propane (Filled) 100 lbs with valve Ref: Specification BB–G–110, Type II	ea	4
257	Cylinder, Oxygen Gas 220–240 cu. ft. (Filled) Ref: Specification BB–O–925 Type I and MIL–C–12661	ea	2
258	Cylinder, Carbon Dioxide (CO_2) Gas, w/valve 50 lb. (Filled) Ref: Specifications MIL–C–12661 and BB–C–101, Grade B, Type II, Class 1	ea	2
259	Cylinder, Gas, Freon 12, 5 lb. container (Filled) Ref: Specifications BB–F–671, Type F–12 and MIL–C–11777	ea	1
260	Cylinder, (b) Marking, Lifetime, Red., Grad., 5 ml. Ref: Specifications DD–V–581, Type V, Class B, Composition a	ea	6
261	Cylinder, (b) Marking, Lifetime, Red., Grad., 10 ml. Ref: Specification DD–V–581, Type V, Class B, Composition a	ea	6
262	Cylinder, (b) Marking, Lifetime Red., Grad., 50 ml. Ref: Specification DD–V–581, Type V, Class B, Composition a	ea	6
263	Cylinder, (b) Marking, Lifetime Red., Grad., 100 ml. Ref: Specification DD–V–581, Type V, Class B, Composition a	ea	24

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TABLE I. *Equipment for One Complete Base Petroleum Laboratory*
(Trade Names and Catalog Numbers are for Reference Only)
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No.	Nomenclature	Unit	Amount required
264	Cylinder, [illegible] Marking, Lifetime Red., Grad., 250 ml. Ref: Spec[illegible]ion DD-V-581, Type V, Class B, Com[illegible]tion a	ea	12
265	Cylinder, (b) Marking, Lifetime Red., Grad., 500 ml. Ref: Specification DD-V-581, Type V, Class B, Composition a	ea	6
266	Cylinder, [illegible]) Marking, Lifetime Red., Grad., 1,000 ml. Ref: Sp[illegible]ification DD-V-581, Type V, Class B, Composition a	ea	6
267	Cylinder, Grad., w/Standard Tapered Stopper, 100 ml. Ref: Specification DD-V-581, Type V, Class B, Composition a	ea	12
268	Cylinder, Grad., w/Standard Tapered Stopper 250 ml. Ref: Specification DD-V-581, Type V, Class B, Composition a	ea	6
269	Desiccator, 250 mm. diameter, w/Porcelain Plate Ref: Specification DD-D-205, Type III, Grade A	ea	4
270	Dish, Evaporating, Nonmetallic, Size 1.90 ml. Ref: Specification NNN-D-340, Type I	ea	6
271	Scriber, Diamond Point, Pencil Writing Ref: Fisher 8-675	ea	2
272	Fexaframe, Set No. 1, Complete Ref: Specification NNN-F-600	ea	2
273	Flask, Erlenmeyer, Pyrex, 50 ml. Ref: Specification NNN-F-240, Type IV, Class I, Style I, Size 3	ea	12
274	Flask, Erlenmeyer, Pyrex, 125 ml. Ref: Specification NNN-F-240, Type IV, Class 1, Style I, Size 4	ea	12
275	Flask, Erlenmeyer, Pyrex, 200 ml. Ref: Fisher 10-040	ea	12
276	Flask, Erlenmeyer, Pyrex, 300 ml. Ref: Fisher 10-040	ea	24
277	Flask, Erlenmeyer, Pyrex, 500 ml. Ref: Specification NNN-F-240, Type IV, Class 1, Style I, Size 6	ea	12
278	Flask, Erlenmeyer, Pyrex, 1000 ml. Ref: Specification NNN-F-240, Type IV, Class 1, Style I, Size 7	ea	6
279	Flask, Iodine, 250 ml. Ref: Specification NNN-F-240, Type IV, Class 2, Style III, Size 2	ea	6
280	Flask, 500 ml. Round Bottom, Short Ring Neck Ref: Specification NNN-F-240, Type I, Class 2, Style II, Size 1	ea	6
281	Flask, Volumetric, 1 liter Cap Stoppered Ref: Specification DD-V-581, Type I, Class B, Composition a	ea	6
282	Flask, Filter, Heavy Wall, 1000 ml. Ref: Specification NNN-F-240, Type V, Size 5	ea	6
283	Funnel, Separatory, 250 ml. Squibb Type Ref: Specification NNN-F-800, Type I	ea	6
284	Funnel, Separatory, 500 ml. Squibb Type Ref: Specification NNN-F-800, Type I	ea	6

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TABLE I. *Equipment for One Complete Base Petroleum Laboratory*
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No.	Nomenclature	Unit	Amount required
285	Funnel, Ribbed Mooney Airvent, 250 ml. Ref: Specification DD-F-796, Class B	ea	6
286	Funnel, Analytical, Pyrex, 75 mm. dia. Ref: Specification DD-F-806, Class B, Size 4	ea	6
287	Furnace, Electric Model TF, 110 Volts, AC Ref: Precision Scientific No. 31000	ea	1
288	System, Gas, Complete with Gage and Connections Ref: Bastian Blessing Co. 4723-A	set	1
289	Gloves, Asbestos Ref: Specification HH-G-450, Type I, Class 1	prs	2
290	Gloves, Rubber, Light Weight, Size 9 Ref: Specification ZZ-G-421, Grade A	prs	6
291	Grease, Stopcock, 1 oz. Tube Ref: Fisher 14-637	tube	6
292	Heaters, Electric Ful Kontrol, 750 W Ref: Precision Scientific No. 61600	ea	4
293	Block, Top, Refractory, upper electric heater Porcelain 3⅛" Hole Ref: Precision Scientific No. 61820	ea	4
294	Heater, Electric Viscosimeter, 110 V., AC Ref: Fisher 13-580	ea	3
295	Hot Plate, 8" Dia., 100 V., AC Ref: Emil Greiner No. G11835-B	ea	3
296	Cylinders (hydrometer Jars) 50 x 350 mm. Ref: Fisher 8-532	ea	6
297	Labels, Gum, Red Border, Size No. 201 Ref: Fisher 11-850	bxs	6
298	Manometer, U-Tube, Gage, Mercury, Pressure or Vacuum Ref: Fisher 11-286	ea	1
299	Mortar, Coors, Porcelain, w/Pestle, Capacity 275 ml. Ref: Specification NNN-M-560, Type I, Class A	ea	2
300	Oven, Electric, 16½ x 16 x 25", 110 V., AC Ref: Precision Scientific Model No. 16, No. 31477	ea	1
301	Paper, Filter, Whatman No. 40, 110 mm. dia. Ref: Fisher 9-845	pkg	2
302	Paper, Filter, Whatman No. 12, 240 mm. dia. Ref: Fisher 9-832	pkg	4
303	Paper, Filter, Heavy White Creped, 33 cm. Ref: Emil Greiner No. G 8328	pkg	2
304	Pencil Scriber, Tungston Carbide Ref: Fisher 13-378	ea	2
305	Pencil, Wax, Red Ref: Fisher 13-380	ea	12
306	Pipette, Transfer, Precision, 50 ml. Ref: Specification DD-V-581, Type III, Class A, Grade 2	ea	6
307	Pipette, Transfer, Precision, 25 ml. Ref: Specification DD-V-581, Type III, Class A, Grade 2	ea	6
308	Pipette, Transfer, Precision, 10 ml. Ref: Specification DD-V-581, Type III, Class A, Grade 2	ea	6
309	Policeman, Scraper, Rubber, Diagonal, Form Ref: MS 36205-1	ea	12

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TABLE I. *Equipment for One Complete Base Petroleum Laboratory (Trade Names and Catalog Numbers are for Reference Only) (or equal) After Each Reference*—Continued

No.	Nomenclature	Unit	Amount required
310	Pump, Vacuum, "Hyvac" w/Iron Base, Serial A, Complete w/Motor, 110 V., 60 Cycle Ref: Cenco 91105	ea	2
311	Plate, Reaction, White, Porcelain, 12 Hole Ref: Specification DD–S–636	ea	2
312	Cork Rings, Suberite, Cork, Inside Dia. 99 mm. Outside Dia. 140 mm. Ref: Specification NNN–R–350, Size 3	ea	6
313	Stoppers, Rubber, 2 Hole, Assorted Sizes from 00–15 Ref: Specification ZZ–S–751, Type B, Grade I	lbs	10
314	Stoppers, Rubber, Solid, Assorted, Sizes from 00–15 Ref: Specification ZZ–S–751, Type A, Grade I	lbs	10
315	Scissors, Dissecting, Medium Weight, Cutting Edge 40 mm. overall length 115 mm. Ref: Fisher 8-945	pr	2
316	Shears, 6" General Utility Ref: Specification GGG–S–278, Type I, Class 1, Style A	pr	3
317	Shears, 12½", General Utility Ref: Specification GGG–S–291, Type II, Class 1	pr	2
318	Sheet, Copper, 12" Wide by 60" Long, 24 Ga Ref: Specification QQ–C–576	sheets	1
319	Rule, Slide, Ortho-Phase, Log-Log Model No. 500 Metal Ref: Chas. Burning Co., New York	ea	2
320	Indicator, Speed, Improved Form Ref: Cenco 73310	ea	1
321	Splashgon Ref: Fisher 9–959	ea	3
322	Sponge, Cellulose, 5¼ x 3½ x 1⅝" Ref: Greene Bros. A 77–220	ea	12
323	Spatula, 4" Blade Ref: Fisher 14–371	ea	6
324	Spatula, 6" Blade Ref: Specification GG–S–593, Class II, Size 3	ea	6
325	Stretcher, Rubber Tubing Ref: Fisher 14–196	ea	6
326	Stopcock, Laboratory, Brass Inlet, Male ⅜" Ref: E. Greiner 22895	ea	24
327	Stirrer, Electric, Fultork, Labmotor, 110 V Ref: Fisher 14–502V2	ea	2
328	Strips, Steel, for Gum Determination Apparatus (100 Strips per pkg.) Ref: E. Greiner No. G 18450	pkg	1
329	Stopcock, Pyrex, Two Way, Straight Bore, 2 mm. Ref: MS 35989–2	ea	6
330	Stands, Tripod, w/Concentric Rings, 9" x 6" Ref: MS 35990–1	ea	4
331	Support Stands, Rectangular Base, Med. Size 5" x 8" Ref: Specification NNN–S–750, Type I, Grade A, Size 2	ea	6
332	Support Stands, Rectangular Base, Large 6" x 9" Ref: Specification NNN–S–750, Type I, Grade A, Size 3	ea	6
333	Rings, Support, Iron, w/Clamp, Size 5" Ref: Specification GG–R–391, Type I, Class A	ea	3
334	Rings, Support, Iron, w/Clamp, Size 4" Ref: Specification GG–R–391, Type I, Class A	ea	3

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TABLE I. *Equipment for One Complete Base Petroleum Laboratory*
(Trade Names and Catalog Numbers are for Reference Only)
(or equal) After Each Reference—Continued

No.	Nomenclature	Unit	Amount required
335	Rings, Support, Iron, w/Clamp, Size 3"	ea	3
	Ref: Specification GG–R–391, Type I, Class A		
336	Tape, Oil Gaging, Steel, Standard, 50' with reel and locking Handle, w/o Bob	ea	3
	Ref: Specification MIL–T–16644		
337	Tag, Shipping, Cloth, w/Wire Fasteners, 4 x 2"	ea	1,000
	Ref: J. J. O'Brien & Son, 123 E 23rd St., New York New York		
338	Thermometer, Cup Case Tank, O–180 F.	ea	4
	Ref: W. Curtin Co., No. 16764A4		
339	Thermometer, General Purpose, —5 to 300 F.	ea	6
	Ref: Cenco 19310–A		
340	Thermometer, General Purpose, —20 to 150 C.	ea	6
	Ref: Cenco 19270–A		
341	Thermo-Regulator, Straight Form Mercury to Platinum Contact	ea	2
	Ref: Precision Scientific No. 62530		
342	Thermo-Regulator Guard	ea	1
	Ref: Precision Scientific No. 62550		
343	Thief, Drum, Oil Resinex	ea	6
	Ref: Emil Greiner No. G–19025		
344	Thief, Sampling, Protek Type, 12" Complete w/16 foot bronze chain	ea	2
	Ref: Lapine No. 335–42 and 335–43A		
345	Tongs, Crucible, 9" CHROME	ea	4
	Ref: MS 36023–2		
346	Tongs, Pickup 7"	ea	2
	Ref: Fisher 15–202		
347	Tongs, Crucible, 18"	ea	1
	Ref: MS 36024–1		
348	Tongs, Beaker, 12"	ea	2
	Ref: MS 36022–1		
349	Forceps, Monel Metal, 6" Gooseneck	ea	2
	Ref: Fisher No. 10–275		
350	Triangle, Chromel Wire, Silica, Type I, Size 1	ea	6
	Ref: Specification NNN–T–685		
351	Triangle, Chromel Wire, Silica, Type I, Size 2	ea	6
	Ref: Specification NNN–T–685		
352	Triangle, Chromel Wire, Silica, Type I, Size 3	ea	6
	Ref: Specification NNN–T–685		
353	Tube, Test, Chemical, w/Rims, Pyrex, 18 x 150 mm	ea	12
	Ref: Cenco 15785		
354	Tubing, Glass, Pyrex OD, 6mm	lbs	10
	Ref: Specification DD–T–741, Type I		
355	Tubing, Glass, Pyrex, OD, 8 mm	lbs	10
	Ref: Specification DD–T–741, Type I		
356	Tubing, Glass, Pyrex, OD, 10 mm	lbs	10
	Ref: Specification DD–T–741, Type I		
357	Tubing, Glass, Pyrex, OD, 12 mm	lbs	10
	Ref: Specification DD–T–741, Type I		
358	Tubing, Rubber, Scimatco, ⅛" Bore Light Wall	ft	25
	Ref: Specification ZZ–T–831, Grade L, Type II		
359	Tubing, Rubber, Scimatco, 3/16" Bore, Light Wall	ft	100
	Ref: Specification ZZ–T–831, Grade L, Type II		

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TABLE I. *Equipment for One Complete Base Petroleum Laboratory*
(Trade Names and Catalog Numbers are for Reference Only)
(or equal) After Each Reference—Continued

No.	Nomenclature	Unit	Amount required
360	Tubing, Rubber, Scimatco, ¼" Bore, Light Wall Ref: Specification ZZ–T–831, Grade L, Type II	ft	50
361	Tubing, Rubber, Scimatco, ⅛" Bore Heavy Wall Ref: Specification ZZ–T–831, Grade L, Type II	ft	25
362	Tubing, Rubber, Scimatco, 3/16" Bore, Heavy Wall Ref: Specification ZZ–T–831, Grade L, Type II	ft	50
363	Tubing, Rubber, Scimatco, ¼" Bore, Heavy Wall Ref: Specification ZZ–T–831, Grade L, Type II	ft	50
364	Tubing, Rubber, Scimatco, 5/16" Bore, Heavy Wall Ref: Specification ZZ–T–831, Grade L, Type II	ft	25
365	Tubing, Pressure and Vacuum, ⅛" Bore Ref: Fisher 14–173	ft	25
366	Tubing, Pressure and Vacuum, 3/16" Bore Ref: Fisher 14–173	ft	25
367	Tubing, Pressure and Vacuum, ¼" Bore Ref: Fisher 14–173	ft	25
368	Watch, Stop, Large 60 Second Dial, Subdivided in ⅕ Seconds, Small Dial Graduated to 30 Minutes in 1 Minute Intervals Ref: Specification GG–S–764, Type I, Class 3	ea	8
369	Watch, Glass, Concave, 35 mm Ref: Specification DD–W–131, Class A	ea	12
370	Watch, Glass, Concave, 65 mm Ref: Specification DD–W–131, Class A	ea	12
371	Watch, Glass, Concave, 100 mm Ref: Specification DD–W–131, Class A	ea	12
372	Distilling Apparatus, Water, Precision, Electrically Heated, 1 Gallon per Hour, 110 Volts, AC Ref: Fisher 9–036	ea	1
373	Automatic Cut-Off (Water Still) Type A, 110 V., AC Ref: Fisher 9–038—5 x 2	ea	1
374	Paste, Water Detector Ref: Emil Greiner No. G 18965	tube	12
375	Wool, Steel, No. 1 Ref: Specification FF–W–556	lbs	3
376	Wool, Steel, No. 0, Medium Ref: Specification FF–W–556	lbs	3
377	Wool, Glass, Filtering Ref: MS 36095	lbs	2
378	Microscope, Chamot, Chemical, Bausch & Lomb, Type L.M. Complete Ref: Fisher 12–333	set	1
379	Calipers, Micrometer, Outside, w/Lock Nut and Ratchet Stop, Complete w/Testing Rods and Case, Starret Co. No. 436D or equal. Range 0 to 6 in. x .001 in, L.S. Starrett Co.	set	1
380	Compound, Grinding and Lapping, Grease Mixed, 4 oz. Can Carborundum No. 277 or equal Ref: C. H. Tiebout & Sons, Inc. 31 Grand St., Brooklyn, New York	ea	6
381	File, Rat Tail, 6" Ref: Specification GGG–F–325, Type XVI, Class 1, Style A	ea	3

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TABLE I. *Equipment for One Complete Base Petroleum Laboratory (Trade Names and Catalog Numbers are for Reference Only) (or equal) After Each Reference*—Continued

No.	Nomenclature	Unit	Amount required
382	File, 3-Cornered, 6" Tapered Ref: Specification GGG–F–325, Type XVIII, Class 2	ea	3
383	Gage, Cylinder, Dial Type, Range 2½" to 6" w/Combination Rigid and Toggle Joint Handle, Locking Mechanism, Starrett No. 452–B or equal, L. S. Starrett Co.	ea	1
384	Gage, Thickness, 26 Leaves, Range .0015 to .025" Ref: Specification GGG–G–17, Type VIII, Class 1, Style A	ea	2
385	Goggles, Safety, Chemical Ref: GGG–G–521, Type II	pr	4
386	Kwickway, CFR Seat Grinder, 110 V. Motor w/Special Small Bearing Grinder Unit for Operation on Special 11/32" Top Arbors, Wheel Dressing Unit and Steel Carrying Case Ref: Cedar Rapids Engr. Co., Cedar Rapids, Iowa Catalog No. 1700	set	1
387	Special Eccentrimeter for Kwickway CFR Grinder	ea	1
388	Special Arbors w/11/32" Top, Size 3/16" for Kwickway CFR Grinder	ea	1
389	Special Arbors w/11/32" Top, Size 5/16" + .005 for Kwickway CFR Grinder	ea	1
390	Special Arbors w/11/32" Top, Size 5/16" + .001 for Kwickway CFR Grinder	ea	1
391	Special Arbors w/11/32" Top, Size 5/16" + .0015 for Kwickway CFR Grinder	ea	1
392	Special Arbors w/11/32" Top, Size 5/16" + .002 for Kwickway CFR Grinder Ref: for above Spare Parts: Cedar Rapids Engineering Company, Cedar Rapids, Iowa	ea	1
393	Arbors, Size 7/16 + .002 for CFR Kwickway Grinder	ea	1
394	All Purpose Grinding Wheels 1 1/16" x 45 degrees for Kwickway CFR Grinder	ea	6
395	All Purpose Grinding Wheel, 1 1/16" x 15 degrees for Kwickway CFR Grinder	ea	6
396	All Purpose Grinding Wheels 1 3/32" x 45 degrees for Kwickway CFR Grinder	ea	6
**397	All Purpose Grinding Wheels 1 3/32" x 15 degrees for Kwickway CFR Grinder	ea	6
398	Flexible Rubber Hose Couplings, for Kwickway CFR Grinder	ea	12
399	Diamond Dressing Tool, for Kwickway CFR Grinder Ref: for Above Spare Parts: Cedar Rapids Engineering Company, Cedar Rapids, Iowa	ea	1
400	Meter, Universal, AC DC Ref: Fisher 9–522	ea	1
401	Oiler, ⅓ Pint, Capacity Ref: GGG–O–591, Type I, Class A	ea	1
402	Puller, 2 Way (Nail) No. 1AVX3 Ref: GGG–P–791, Type I, Class 2	ea	1
403	Screw Driver, 3" Blade Standard GGG–S–121, Type I, Class 5, Style 1, Design A, Shape A	ea	2

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TABLE I. *Equipment for One Complete Base Petroleum Laboratory (Trade Names and Catalog Numbers are for Reference Only) (or equal) After Each Reference*—Continued

No.	Nomenclature	Unit	Amount required
404	Screw Driver, 6" Blade Standard GGG–S–121, Type I, Class 5, Style 1, Design A, Shape A	ea	2
405	Solder, Resin Center, 1 lb. Spools Ref: Greybar Electric Co.	lbs	2
406	Solder Acid Core, 1 lb. Spools Ref: Greybar Electric Co.	lbs	2
407	Stool, Laboratory, Steel, 30" Tall See Drawing #5–14–322	ea	4
408	Wrench, Pipe, Adjustable 6" Standard GGG–W–651, Type I, Class A	ea	2
409	Wrench, Pipe, Adjustable 8" Standard GGG–W–651, Type I, Class A	ea	2
410	Wrench, Pipe, Adjustable 10" Standard GGG–W–651, Type I, Class A	ea	2
411	Wrench, Pipe, Adjustable 14" Standard GGG–W–651, Type I, Class A	ea	1
*412	Wrench, Pipe, Adjustable 18" GGG–W–651, Type I, Class A Standard	ea	1
413	Wrench, Crescent, Adjustable, 6" Standard GGG–W–631, Type I, Class 1	ea	2
414	Wrench, Crescent, Adjustable, 8" Standard GGG–W–631, Type I, Class 1	ea	2
415	Wrench, Crescent, Adjustable, 10" Standard GGG–W–631, Type I, Class 1	ea	2
416	Wrench, Crescent, Adjustable, 12" Standard GGG–W–631, Type I, Class 1	ea	1
417	Wrench, Crescent, Adjustable, 15" Standard	ea	1
418	Iron, Soldering, Electric, ½ lb Ref: Cenco 88765	ea	1
*419	Iso-Octane, w/Blending Tables, 55 Gal. Drums Ref: Standard Oil Co. of N. J. (ENJAY)	Drums	2
*420	Normal Heptane, w/Blending Tables, 55 Gal. Drums Ref: Phillips Petroleum Company, Bartletsville, Okla.	Drums	2
*421	Benzene, Nitration, Grade, ASTM D835–47 in 55 Gal Drum Ref: Barrett Co., 40 Rector St., New York, N. Y	Drum	1
*422	Reference Fuel Grade Toluene in 55 Gallon Drum Ref: Phillips Petroleum Company	Drum	1
423	Aprons, Rubber, Length 46" Width 36" Ref: E. Greiner Co. G–875	ea	6
424	Union Nuts, Tube OD Size ⅜" #W41 x 6 Standard, Weatherhead Company	ea	6
425	Elbow Union, Tube OD ⅜" # W 55 x 6 Standard, Weatherhead Company	ea	6
426	Elbow Male, Tube OD ⅜" # W 49 x 6 Standard, Weatherhead Company	ea	6
427	½" Union, Tube OD ⅜" # W 48 x 6 Standard, Weatherhead Company	ea	6
428	3-Way Teel Union, Tube OD ⅜" # W 44 x 6 Ref: Weatherhead Company	ea	6
429	3-Way Tee Union, Tube OD ⅜" # W 51 x 6 Ref: Weatherhead Company	ea	6

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TABLE I. *Equipment for One Complete Base Petroleum Laboratory (Trade Names and Catalog Numbers are for Reference Only) (or equal) After Each Reference*—Continued

No.	Nomenclature	Unit	Amount required
430	Valve, Globe, Brass ¼" Standard, Ref: Crane Company	ea	6
431	Valve, Globe, Brass ⅜" Ref: Crane Company 47–30 29th St., Long Island, N.Y.	ea	6
432	Hood, Chemical Fufe, Complete with Blower (See Drawing # 5–14–335) Ref: Equipment & Furniture Company	set	1
433	Sink, Laboratory with Double Drain Board (see Drawing # 5–14–333) Ref: Equipment & Furniture Company	ea	1
434	Visconimeter, Kinematic, Low Temperature, Refrigerated, Electric (See Drawing # 5–14–336) Ref: Lawler Electric Manufacturing Co.	ea	1
435	Compressor, Air, Ingersoll-Rand, 2 Stage Unit, Type 30; 200 lbs. p.s.i. Continuous Service, one (1) Horse-Power, Tank Size 20" x 50", 60 Gallon, 60 Cycles, 110/220 V, Single Phase, A.C.	ea	1
436	Switch, Motor Protection for Air Compressor, Single Phase, 110 Volts Ref: Renich & Mahoney, Inc. Catalog #232 C–1	ea	1
437	Refrigerator, Electric (see Drawing #5–14–337) Ref: Lawler Electric Mfg. Company	ea	1
438	Cloud and Pour Point Apparatus (see Drawing # 5–14–338)	ea	1
439	Channel Point Apparatus (see Drawing # 5–14–340) Ref: Lawler Electric Mfg. Company	ea	1
440	Regulator, Oxygen, Complete with High and Low Pressure Gages** Ref: National Cylinder Gas Company	ea	1
441	Barette, Reference Fuel Blending (Octane), 400 ml Capacity** Ref: Scientific Glass Apparatus Company, Bloomfield, N.J.	ea	4
442	Freezing Point Apparatus, Gasoline Ref: Emil Greiner Co., No. G 17535	ea	2
444	Foam Test Apparatus, Complete ASTM, D892 Ref: E. Greiner Co. G 18251	set	1
445	Sampling Device, ASTM, 1 Qt. Capacity, ¾" Opening Ref: Precision #74856	ea	4
446	Strip, Copper, ½" x 3" x ⅛" Ref: MS36252	ea	12
447	Standards, Corrosion, Copper ASTM Method 5313 of Federal Specification VV–L–791e Ref: E. Greiner Co., No. G 17757	set	1
448	Bob, Plumb, Brass, Outage 16 oz. Ref: Lufkin Rule Co. 587	ea	3
449	Bob, Plumb, Brass, Innage 20 oz.	ea	3
450	S.O.D. Lead Corrosion Apparatus Complete with Constant Temperature Bath (Liquid Bath) 6 Unit, capable of maintaining a uniformed temperature of 325 F. ± 2 F. (163 ± 1 C.) throughout the bath. The bath having provisions for 6 sample tubes. A vertical support houses 6 stainless steel stirring shafts driven by an electric motor capable of rotating the shafts at a speed of 600 ± 5 r.p.m. Six flow-		

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TABLE I. *Equipment for One Complete Base Petroleum Laboratory (Trade Names and Catalog Numbers are for Reference Only) (or equal) After Each Reference*—Continued

No.	Nomenclature	Unit	Amount required
	meters calibrated for 2 cubic feet of air per hour complete but without sample tubes or test panels for operation on 220 volts 50/60 cycle A.C. 6 complete with S.O.D. sample tubes for both. 100 Test Panels Lead (4 lb chemical lead) 1⅜ x 1⅝" with 3/16" hole in center of panel. 100 Test Panels Copper (QQ–C–576) 1⅜ x 1⅝" with 3/16" hole in center	set	1
	Ref: As described in Specification MIL–L–780B		
**451	Grid, Wire Gauze, 10 mesh, w/3" asbestos center patch, size 5" square	doz	1
	Ref: E. Greiner No. G 26135		
**452	Flexible Metal Braided Hose, ½" I.P.S.	ea	30
**453	Flexible Metal Braided Hose, 1" I.P.S.	ea	9

*Not to be included in cabinet assembly, but to be supplied by contractor.
**Storage space not provided. To be located at the convenience of contractor.

TABLE II. *Chemicals Required For One Base Petroleum Laboratory Assembly*

No.	Nomenclature	Unit	Amount required
601	Acetic, Acid, Glacial, 99.7% ACS	pt	2
	Ref: Specification O–C–265		
602	Acetone, USP, Tech	ga	5
	Specification O–C–265		
603	Alkacid Tester, Fisher PH	ea	5
	Ref: Fisher 14–834–1*		
604	Ammonium Acetate, ACS, MW 77–08	lb	1
	Ref: Specification O–C–265		
605	Ammonium Chloride, ACS, MW 53.50	lb	1
	Ref: Specification O–C–265		
606	Ammonium Hydroxide, ACS Sp. Gr. 0.900	lb	8
	Ref: Specification O–C–265		
607	Ammonium Iodide, N.F.	lb	¼
	Ref: Fisher A–937		
608	Ammonium Molybdate, ACS MW 1235.95	lb	1
	Ref: Specification O–C–265		
609	Ammonium Nitrate, ACS	lb	¼
	Ref: Specification O–C–265		
610	Ammonium Oxalate, ACS	lb	1
	Ref: Specification O–C–265		
611	Ammonium Sulfate, ACS	lb	1
	Ref: Specification O–C–265		
612	Anhydrous Isopropyl Alcohol	gal	1
	Ref: Specification O–C–265		
613	Aniline, ACS, MW, 93.12	pt	3
	Ref: Specification O–C–265		
614	Asbestos, Fiber, Acid Washed, for Gooch Crucible	lb	1
	Ref: Fisher 1–458–6		
615	Barium Chloride, Anhydrous, FW 208.27	lb	1
	Ref: Specification O–C–265		

*All references are Fisher Scientific Company.

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TABLE II. *Chemicals Required For One Base Petroleum Laboratory Assembly*—Continued

No.	Nomenclature	Unit	Amount required
616	Barium Hydroxide A.C.S. Ref: Specification O–C–265	lb	¼
617	Benzone (Benzol) ACS, Sp. Gr. 0879 Ref: Specification VV–B–231, Grade A	gal	5
618	Benzoic Acid, NBS Certificate Ref: Fisher 4–460	grms	60
619	Bromine, ACS, At. W. 79.92 Sp. Gr. 3.119 Ref: Specification O–C–265	oz	4
620	Calcium Chloride, Calcine, Porous, Tech. 8 mesh Ref: Specification O–C–105 Type II, Grade B, Class 1	lb	25
621	Carbon, Tetrachloride, Tech Ref: Specification O–C–265	gal	5
622	Celite (Analytical Filter Aid) Ref: Fisher C-211	lb	1
623	Chloroform, ACS Ref: Specification O–C–265	lb	1
624	Cobaltous Chloride, Crystal Ref: Specification O–C–265	lb	¼
625	Cupric Sulfate Powder, Technical Ref: Fisher C–490	lb	1
626	Ethyl Alcohol, 95% ACS Ref: Specification O–E–760, Grade I, Class B	gal	5
627	Ethylene Glycol Purified Ref: Fisher E–177	gal	5
628	Ferric Choloride, ACS Ref: Specification O–C–265	lb	¼
629	Glycerin, White, USP Ref: Fisher G–32	lb	1
630	Hydrochloric Acid, ACS Sp. Gr. 1.19 Ref: Specification O–C–265	lb	30
631	Hydrofluoric Acid, 48% ACS Ref: Specification O–C–265	lb	¼
632	Hydrogen Peroxide, Tech 30–35% Ref: Fisher H–327	lb	10
633	8–Hydroxyguinoline (ortho) ACS Ref: Specification O–C–265	lb	¼
634	Iodine Resublimed, ACS Ref: Fisher 1–37	lb	1
635	Ferrous Sulfide Granular Ref: Fisher I–150	lb	1
636	Lead Acetate, ACS Normal MW 379.35 Ref: Specification O–C–265	lb	1
637	Lead Oxide, Mono (Litharge) PBO Ref: Fisher L–68	lb	1
638	Mercury Metal, Sp. Gr. 13.546 Ref: MIL–M–191, Grade II	lb	5
639	Methanol ACS Ref: Specification O–C–265	gal	5
640	Methyl Ethyl Ketone, Tech Ref: Specification TT–M–261	gal	5
641	Methyl Orange (indicator) ACS Ref: Fisher M–216	oz	8
642	Methyl Red (Indicator) ACS Ref: Fisher M–219	lb	¼
643	Nitric Acid, Sp. Gr. 1.42, ACS Ref: Specification O–C–265	lb	21

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TABLE II. *Chemicals Required For One Base Petroleum Laboratory Assembly*—Continued

No.	Nomenclature	Unit	Amount required
644	Naphtha Solvent, ASTM D–91–52 Ref: Fisher N–3	gal	5
645	Xylene (NM) practical Ref: Fisher 275–P	gal	5
646	Naphtha Solvent, Zylene, D–95–46 Ref: Fisher X–4	gal	5
647	Oleic Acid, Tech Ref: Fisher A–215	lb	1
648	P–Nitrophenol, Reagent Ref: Fisher N–105	grms	100
649	Phenolphthalein, ACS Ref: Specification O–C–265	oz	8
650	Petroleum Ether, ACS Ref: Specification O–C–265	gal	5
651	Phosphorous Pentoxide, Anhydride, ACS MW 141.96 Ref: Specification O–C–265	lb	1
652	Potassium Bromide, Crystal ACS Ref: Specification O–C–265	lb	¼
653	Potassium Chloride, ACS, MW 74.55 Ref: Specification O–C–265	lb	1
654	Potassium Chromate, ACS, MW 194.20 Ref: Specification O–C–265	lb	5
655	Potassium Dichromate, Crystal, ACS, MW 294.21 Ref: Specification O–C–265	lb	10
656	Potassium Ferrocyanide, Crystal, ACS– MW 422.39 Ref: Specification O–C–265	lb	1
657	Potassium Hydroxide, Pellets, ACS, MW 56.10 Ref: Specification O–C–265	lb	5
658	Potassium Iodide, Granular, ACS, MW 166.02 Ref: Specification O–C–265	lb	1
659	Potassium Nitrate ACS fine granular Ref: Specification O–C–265	lb	¼
660	Potassium Permanganate ACS Crystal Ref: Specification O–C–265	lb	¼
661	Silver Nitrate Crystal ACS Ref: Specification O–C–265	oz	8
662	Sodium Carbonate, Anhydrous, ACS MW 106.00 Ref: Specification O–C–265	lb	10
663	Sodium Chloride White, Tech Ref: Fisher S–269	lb	10
664	Sodium Hydroxide, Pellets, ACS, MW 40.01 Ref: Specification O–C–265	lb	10
665	Sodium Nitrate Crystal, ACS, MW 85.01 Ref: Specification O–C–265	lb	1
666	Sodium Oxalate, MW 134.004 ACS Ref: Specification O–C–265	lb	1
667	Sodium Peroxide, 20 mesh MW 77.99 Ref: Specification O–C–265	lb	1
668	Sodium Sulfate, Anhydrous, ACS, MW 142.05 Ref: Specification O–C–265	lb	2
669	Sodium Thiosulfate, Crystal ACS, MW 248.194 Ref: Specification O–C–265	lb	1
670	Starch, Corn Powder, USP Ref: Fisher S–510	lb	2
671	Stoddard Solvent Ref: SpecificationP–S–661, Type I	gal	5

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TABLE II. *Chemicals Required For One Base Petroleum Laboratory Assembly*—Continued

No.	Nomenclature	Unit	Amount required
672	Sulfur, sublimed (Sulfur Flower) Ref: Fisher S–591	lb	1
673	Sulfuric Acid, Sp. Gr. 1.84, ACS, MW 98.08 Ref: Specification O–C–265	lb	36
674	Tannic Acid F. W. 322.234 Ref: Powder Form Fisher No. A–308	lb	1
675	Tripoli Powder Ref: Fisher T–358	lb	5
676	Zinc Oxide, ACS Ref: Specification O–C–265	lb	¼

TABLE III. *Waukesha Equipment and Parts For Base Petroleum Laboratory Assembly*

Waukesha No.	Nomenclature	Unit	Quantity required
701 Model V 16	Engine, Fuel Testing, C.O.T. Military, Motor and research methods complete	ea	1
702 B–2557	Gasket, Carburetor and Exhaust Manifold	ea	2
703 B–2676	Tube, Neon, Spark Indicator	ea	2
704 B–3207	Washer, Carburetor Horizontal Jet Hole	ea	2
705 B–3663	Nut, ⅛" SAE Flared, Pressure Line	ea	1
706 B–3665	Fitting, ⅛" Straight	ea	1
707 B–5052	Washer, Carburetor Float Gage Glass, Top	ea	4
708 B–5052	Washer, Carburetor Fuel Tank (Lower)	ea	2
709 B–5052–A	Washer, Condenser Gage, Lower	ea	2
710 B–5068	Washer, Carburetor Float Gage Glass, Bottom	ea	2
711 B–5068	Washer, Condenser Gage & Fuel Tank Gage (Upper)	ea	2
712 B–5094	Glass, Carburetor Fuel Tank Gage	ea	2
713 B–5096–A	Gasket, Water Pipe	ea	2
714 24088	Wrench, Crank Gear Nut	ea	1
715 24088–S	Wrench, Spark Plug	ea	1
716 24088–T	Wrench, Fly Wheel Nut	ea	1
717 27736	Voltmeter	ea	1
718 27827	Wrench, Adjustable End (6 inch)	ea	1
719 27828	Wrench, Box Socket, 11/16" x ¾" Double	ea	1
720 27829	Wrench, Box Socket, ½" x 9/16" Double End	ea	1
721 27830	Wrench, Box Socket, ⅜" x 7/16" Double End	ea	1
722 27831	Screw Driver	ea	1
723 75552–A	Gasket, Carburetor Fuel Tank Gage	ea	2
724 75623	Coil, Ignition	ea	1
725 75690	Gasket, Condenser Body	ea	2
726 75775–A	Stone, Breaker Point	ea	1
727 75974	Glass, Carburetor Float Gage	ea	2
728 75977	Cap, Carburetor Float Gage	ea	3
	Jets, Carburetor, Fuel, Horizontal, C.O.T. Engine, as follows:		
729 75988–A	50 CC	ea	3
730 75988	52 CC	ea	3
731 75988	54 CC	ea	3
732 75988	56 CC	ea	3
733 75988	58 CC	ea	3
734 75988	60 CC	ea	3
735 75988	62 CC	ea	3

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TABLE III. *Waukesha Equipment and Parts For One Base Petroleum Laboratory Assembly*—Continued

Waukesha No.	Nomenclature	Unit	Quantity required
736 75988	64 CC	ea	3
737 75988	66 CC	ea	3
738 75988	68 CC	ea	3
739 75988	70 CC	ea	2
740 78154	Plug, Spark	ea	2
741 105180	Thermometer, Oil, Water, Air, Range 0–220 F.	ea	2
742 105267	Ammeter	ea	1
743 105321–D	Gage, Oil Temperature	ea	1
744 105450	Wrench, Handle	ea	1
745 105599	Fuse, Plug, Panel	ea	1
746 105991	Gasket, Manifold	ea	2
747 106317–A	Thermometer, Air Intake, Range 60–160 F.	ea	2
748 106747–A	Manifold Heating Housing, Flange Type	ea	1
749 106820	Thermometer, Heater, Range 300–400 F.	ea	2
750 108288	Valve, Intake, Shrouded	ea	1
751 108289	Valve, Exhaust	ea	1
752 108295	Ring, Piston, Compression (Tapered)	ea	3
753 108296	Ring, Piston, Oil (Square)	ea	4
754 108297	Ring, Piston, Oil (Tapered)	ea	2
755 109221	Burette	ea	1
756 109284	Gage, Feeler	ea	1
757 109346	Gasket, Carburetor, Thick, 2 Bolt	ea	2
758 109421	Gasket, Water Inlet Cap, Coolant Drainpipe	ea	2
759 109802	Screwdriver, Allen	ea	1
760 109885	Pliers, Piston-Pin Retaining Ring	ea	1
761 210124	Fuse, Detonation Meter	ea	1
762 0105976–A	Assembly, Condenser Gage Level	ea	1
763 0109418	Cylinder Pipe	ea	1
764 W109664	Pack, Power	ea	1
765 W109926	Meter, Detonation, Model 501–A, W/Cable Assy.	ea	1
766 W109927	Pick-up, Detonation	ea	1
767 0110351	Thermometer, Intake Manifold 200–350 F.	ea	2
768 Special	Expander, Piston Ring, for C.O.T. Engine	ea	1
769	Tachmometer, Mechanical, Hand Held Chronometric Type, 10 Second Timing Interval Range 0–10,000 RPM, 10 RPM Smallest Increment 0–3,000 RPM, 2 Pointers; Manual Reset. Complete W/Carrying Case. MIL–T–16040 Type IC/APA	ea	1

TABLE IV. *Text Books Required For Laboratory Petroleum Base*

No.	Nomenclature	Unit	Amount required
801	Federal Test Method Standard No. 791, Lubricants; Liquid Fuels, and Related Products; Method of Testing (see 2.1). Ref: GSA, Washington, D.C.	ea	4
802	ASTM Standards on Petroleum Products and Lubricants Latest Edition Ref: American Society for Testing Materials.	ea	1
803	New and Revised Tag Manual for Inspectors of Petroleum Ref: Fisher Scientific Co., Latest Edition	ea	2

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TABLE IV. *Text Books Required For Laboratory Petroleum Base*—Continued

No.	Nomenclature	Unit	Amount required
804	Handbook of Chemistry and Physics-Latest Edition Ref: Chemical Rubber Publishing Company, 2310 Superior Avenue, N.E., Cleveland, Ohio	ea	1
805	Lubrication of Ordnance Material Ref: Department of the Army, Ordnance Department, Washington, D.C.	ea	1
806	Booklet of National Petroleum Correction Tables Ref: Bureau of Standards, Supt. of Documents Washington, D.C.	ea	1
807	The Significance of Test of Petroleum Products Ref: ASTM (see item 2)	ea	1
808	Technical Methods of Analysis—Latest Edition Ref: McGraw-Hill Book Company, Inc., New York, N.Y.	ea	1
809	Instructions for Measuring, Sampling and Testing Petroleum Shipments Ref: Standard Inspection Laboratory, Standard Oil Development Company, New York, N.Y.	ea	1
810	Chemical Engineers Handbook by Perry Ref: McGraw-Hill Book Company, Inc., New York, N.Y.	ea	1
811	ASTM Manuel, Engine Test Methods for Rating Fuel by Supercharged, Motor, Research Aviation & Cetane Methods—Latest Edition Ref: American Society for Testing Materials, Philadelphia, Pa.	set	1
812	Chemistry of Petroleum by Ellis—Volume 1 Ref: Acme Code Company, 89 Broad Street, New York, N.Y.	ea	1
813	Chemistry of Petroleum by Ellis—Volume II Ref: Acme Code Company, 89 Broad Street, New York, N.Y.	ea	1
814	Evaluation of Petroleum Products Ref: American Society for Testing Materials, Philadelphia, Pa.	ea	1
815	Scotts Standard Methods of Chemical Analysis—Vol. 1 Ref: Acme Code Company, 89 Broad Street, New York, N.Y.	ea	1
816	Scotts Standard Methods of Chemical Analysis—Vol. II Ref: Acme Code Company, 89 Broad Street, New York, N.Y.	ea	1
817	Operating, Wiring Diagram and Shipping Instructions (1 set on each refrigeration unit and 1 set in envelope in bookcase.)	ea	2

TABLE V. *Additional Items For Air Force Procurement*

No.	Nomenclature	Unit	Amount required
901	Cone, wire gauze Ref: Emil Greiner Co., No. G8281	ea	1
902	Beaker, 200 ml with cover assembly Ref: Emil Greiner Co., No. G1711	ea	6
903	Template, Dirt Content of Grease Ref: Emil Greiner Co., No. G30051	ea	1

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TABLE V. *Additional Items For Air Force Procurement*—Continued

No.	Nomenclature	Unit	Amount required
904	Chamber, Vacuum with removable cover and vacuum connection Ref: Emil Greiner Co. No. G30051–A	ea	1
905	Bath, Air-Well, 4 unit complete with cover and constant level device Ref: Emil Greiner Co., No. G5313	ea	1
906	Rack, Air-well bath, adjustable Ref: Emil Greiner Co., No. G5313A	ea	1
907	Tube, test, flat, for copper strips Ref: Emil Greiner Co., No. G19240	ea	12
908	Thermometer ASTM, Gravity minus 5 to plus 215 F. Ref: Emil Greiner Co., No. G23594	ea	6
909	Holder, Copper Strip, single unit Ref: Emil Greiner Co., No. G5313B	ea	1
910	Column, absorption, standard Ref: Emil Greiner Co., No. G18246	ea	1
911	Tubing, analyser, 3 mm. OD x 1.5 mm bore in 4 ft. lengths Ref: Emil Greiner Co., No. G18246D	lb	1
912	Lamp, Ultra Violet, single tube, 49 inches long, 3650 Angstrom units Ref: Emil Greiner Co., No. G18246F	ea	1
913	Vibrator, Packing Column Ref: Emil Greiner Co., No. G18246M	ea	1
914	Titrimeter, Dual, Precision shall complete with meter and two titration assemblies Ref: E. Greiner Co., No. G17300	ea	1
915	Explosimeter, complete Ref: Emil Greiner Co., No. G11511	ea	1
916	Thermometer range 100—150 F. Ref: Emil Greiner Co., No. G11511D	ea	1
917	Bottles, Test A & B, complete with glass and copper tubes and stoppers Ref: Emil Greiner Co., No. G11511C	set	1
918	Flask, Volumetric, 2000 ml, fitted with rubber Stopper, glass tubes and pinch clamps Ref: Emil Greiner Co., No. G11511A	ea	2
919	Cell Evaporation, complete Ref: E. Greiner Co., No. G295-1	ea	1
920	Bath, oil, constant temperature suitable for immersion of evaporation cell for tests at 2.0 to 300 F. Ref: Emil Greiner Co., No. 295–2A	ea	1
921	Flowmeter, for evaporation cell Ref: Emil Greiner Co., No. 295–2	ea	1
922	Bath, constant temperature, Aluminum Block, 6 unit for air at 325 F. or super heated steam at 455 F. complete with built-in super-heater and automatic control Ref: Emil Greiner Co., No. G18416	ea	1
923	Gauge, Vapor Pressure, Range 0–5 pounds in 1/10 lb. division Ref: E. Greiner Co., No. G19504	ea	2
924	Cylinder, Mixing, Single, Graduated, Standard taper, glass stoppered, 100 ml Ref: MS 35947–6	ea	48
925	Thermometer set, Anschutz with National Bureau of Standards Certificate, complete with case Ref: E Greiner No. G24122	set	1

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TABLE V. *Additional Items For Air Force Procurement*—Continued

No.	Nomenclature	Unit	Amount required
926	Silica carbide paper 240–250 grit, No. G5313C Ref: Emil Greiner Co.	ea	24
927	Silica carbide, powder, 150 mesh, No. G5313D Ref: Emil Greiner Co.	lb	1
928	Silica Gel 100–200 mesh, No. G7688K	lb	1
929	Indicator, Dye Fluorescent, 10cc per vial for Fluorescent Indicator Absorption F.I.A. vial Ref: Emil Greiner Co., No. G18246R	ea	6
	Contractor to locate the above items in drawers and cabinets whenever possible.		

TABLE VI. *Packing List*

Item No.	Description	Quantity
	Unit No. 1—Laboratory Petroleum Base: One Armorply Shipping Crate and Platform. One Enamel Steel Cabinet. One Elevating Platform. One Stainless Steel Cover. One sliding Pipe Rack, Complete.	
157	Bath, Water, Thermostatically Controlled	1
157A	Tube, Overflow, Component part, Water Bath	1
197	Burner, Gas	2
198	Burner, High Temperature	2
199	Burner, Pittsburgh, Universal	2
201	Burner, Micro	1
202	Burner, Forbes, Bunsen	2
300	Oven, Electric, Overall dimensions 16½" wide 16" deep and 25" high	1
358	Tubing, Rubber, Scimatco, ⅛" Bore, Thin Wall	25 Ft.
359	Tubing, Rubber, Scimatco, 3/16" Bore, Thin Wall	100 Ft.
361	Tubing, Rubber, Scimatco, ⅛" Bore, Thick Wall	25 Ft.
366	Tubing, Pressure and Vacuum, 3/16" Bore, Thick Wall	25 Ft.
367	Tubing, Pressure and Vacuum, ¼" Bore, Thick Wall	25 Ft.
407	Stool, St. St., Collapsible	1
452	Flexible, Metal Braided Hose ½" I.P.S.	1
365	Tubing, Pressure and Vacuum, ⅛" Bore	25 Ft.
	Unit No. 2: One Armorply Shipping Crate and Platform. One Enameled Steel Cabinet. One Elevating Platform. One Stainless Steel Cover. One Sliding Pipe Rack, Complete.	
124	Pan with Lip, 1 Qt. Cap Viscosimeter	1
125	Strainer, Oil, Viscosity	2
343	Thief, Drum, Oil Resinex	1
115A	Bulb #NE 30 (Vapor Pressure App.) Component Part	1
123	Kinematic Viscosity Bath	2
123A	Kinematic Viscosity Control Box, Component Part	2
123B	Kinematic Viscosity Relay, Component Part	2
442	Freezing Point Apparatus, Gasoline	2
452	Flexible, Metal Braided Hose, ½ inch I.P.S.	2

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TABLE VI. *Packing List*—Continued

Item No.	Description	Quantity
	Unit No. 3:	
	One Armorply Shipping Crate and Platform.	
	One Enameled Steel Cabinet.	
	One Elevating Platform.	
	One Stainless Steel Cover.	
	One Sliding Pipe Rack, Complete.	
28	Distillation Apparatus, Front View w/750 W., Heater; 115 V., Gasoline; complete	2
	Ref: Greiner G17880	
28A	Stand, Round, Wood, Component Part	3
28B	Refractory Block, Ceramic, Component Part	6
28C	Drain Cock, ⅛ inch Female, Component Part	3
28D	Overflow, ⅛ Inch Male, Component Part	3
28E	⅛ inch Pipe Drain Line, Component Part, Distillation Apparatus	4
452	Flexible Metal Braided Hose ½ inch, I.P.S.	1
171	Bottle, Narrow Mouth, Clear, Round, 1 Qt. Cork Tip	1
	Unit No. 4:	
	One Armorply Shipping Crate and Platform.	
	One Enameled Steel Cabinet.	
	One Elevating Platform.	
	One Stainless Steel Cover.	
	One Sliding Pipe Rack, Complete.	
38	Flack, Distilling, 100 cc	14
28	Distillation Apparatus, Front View	1
28E	⅛ inch Pipe Drain Line, Component Part, Distillation Apparatus	2
39	Flask, Distilling, 250 cc	8
33	Cylinder, Single Grad., Glass 200 ml 0–100% Special	6
34	Jar. Battery, 4⅝ inch dia. by 9 inch high for Dist. Apparatus	4
452	Flexible Metal Braided Hose, ½ inch I.P.S.	1
	Unit No. 5:	
	One Armorply Shipping Crate and Platform.	
	One Enameled Steel Cabinet.	
	One Elevating Platform.	
	One Stainless Steel Cover.	
	One Sliding Pipe Rack, Complete.	
90	Cone, Grease	2
89	Penetrometer, Complete	1
92	Needle, Penetrometer	2
103	Smoke Pt. Apparatus	1
104	Wick for Smoke Pt. Apparatus	1
112	Stand, Sulfur Determination Apparatus	2
115B	Merc To Merc Relay (Vapor Pressure App.) Component part	1
115B	Rod and Carriage (Vapor Pressure App.) Component part	1
244	Cork, Press, Rotary	1
287	Furnace, Elec. Model, T.F.110V	1
287A	Pan for Furnace, Component part	1
299	Mortar, Coors, Porcelain, w/Pestle, Size #1	1
311	Plate, Reaction, White, Porcelain, 12 Holes	2
407	Stool, St. St., Collapsible	2
452	Flexible, Metal Braided Hose ½ inch I.P.S.	1
	Unit No. 6:	
	One Armorply Shipping Crate and Platform.	
	One Enameled Steel Cabinet.	
	One Elevating Platform.	
	One Stainless Steel Cover.	
	One Sliding Pipe Rack, Complete.	

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TABLE VI. *Packing List*—Continued

Item No.	Description	Quantity
51	Gum Content of Gasoline Appar. ASTM, 110V	1
294	Heater, Elec., Viscosimeter, 110V	3
337	Tag, Shipping Cloth, w/Wire Fasteners 4 inch by 2 inch	1,000
374	Paste, Water Detector	12
298	Gage, Manometer, Mercury, Pressure or Vac	1
336	Tape, Oil Gaging, Steel, Standard 50 Ft. with reel and locking Handle w/o Bob, Specification MIL–T–16644	3
338	Thermometer, Cup Case Tank, 0–180 F	4
344	Thief, Sampling, Tulsa Type, 12 inch	2
452	Flexible, Metal Braided Hose, ½ Inch I.P.S.	1
46 & 49	Ferrels (P.M. & Tag) Component	6
123	Metal Stirrer (Vis) Component	2
	Unit No. 7: One Armorply Shipping Crate and Platform. One Enameled Steel Cabinet. One Elevating Platform. One Stainless Steel Cover. One Sliding Pipe Rack, Complete.	
91	Grease Worker	1
178	Bottle, Narrow Mouth, 5 Gal. Cap. Clear, Round	1
372	Distilling Apparatus, Water, Precision Elec. Heated, 1 Gal./Hr., 110 V	1
373	Automatic Cutoff (Water-Still) Type A, 110 V	1
407	Stool, St. St., Collapsible	1
452	Flexible Metal Braided Hose, ½ inch I.P.S.	1
	Unit No. 8: One Armorply Shipping Crate and Platform. One Enameled Steel Cabinet. One Sliding Pipe Rack, Complete. One Enameled Steel Bookcase.	
107	Burner for Sulfur Determination Apparatus	2
109	Flask, Lamp, Pyrex, Sulfur Deter. Apparatus	5
113	Tetraethyl Lead Extraction Apparatus, Complete	3
95	Flask, Wide Mouth, 1,000 ml	1
118	Vapor Pressure Bomb, Complete	3
148	Flask, Pyrex, 500 ml. Round Bottom, Long Neck	1
108	Chimney, Pyrex, Sulfur Deter. Apparatus	2
110	Spray, Trap, Ground Joint	2
106	Absorber, Pyrex Sulfur Deter. Apparatus	2
111	Wicks for Sulfur Deter. Apparatus	144
113A	Rheostats for T.E.L. Ref: Component part.	3
105	Sulfur Determination Apparatus, Complete	2
310	Pump, Vacuum "Hyvac" w/Iron Base, Serial "A" w/Motor, 110 V., 60 cyc	1
452	Flexible Metal Braided Hose ½ inch I.P.S.	1
802	ASTM Standards on Petr'l, Products and Lubricants	1
803	New and Revised Tag Manual for Inspectors of Petr'l	2
804	Handbook of Chemistry and Physics—Latest Edition	1
805	Lubrication of Ordnance Material	1
806	Booklet of National Petr'l Correction Tables	1
807	The Significance of Test of Petr'l Products	1
808	Technical Methods of Analysis	1
809	Instructions for Measuring, Sampling and Testing Petr'l Shipments	1
810	Chemical Eng. Handbook	1
811	ASTM Manual Engine Test Methods for Rating Fuel	1
812	Chemistry of Petroleum—Volume 1	1

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TABLE VI. *Packing List*—Continued

Item No.	Description	Quantity
813	Chemistry of Petroleum—Volume 2	1
814	Evaluation of Petr'l Products	1
815	Scott's Standard Methods of Chemical Analysis—Vol 1.	1
816	Scott's Standard Methods of Chemical Analysis—Vol. 2	1
	Unit No. 9: One Armorply Shipping Crate and Platform. One Enameled Steel Cabinet w/s. s. top. One Sliding Pipe Rack, Complete.	
3	Thermometer, ASTM, Analine Pt. App. 25–105 degrees C	6
35	Thermometer, 30–580 degrees F., Dist. App	12
36	Thermometer, 30–760 degrees F.	6
43	Thermometer, 20–580 degrees F., Dropping Pt	6
45	Thermometer, 20–760 degrees F., Flash Pt	6
47	Thermometer, ASTM, 20–230 degrees F., Pensky Marten	6
48	Thermometer, 200–700 degrees F	4
50	Thermometer, Closed Tester, —4 to 122 degrees F	6
58	Thermometer, 204–218 degrees F., Gum Stab. App	6
101	Thermometer 0–220 degrees F., Steam Emulsion	4
102	Thermometer, 30–230 degrees F., Steam Emulsion	4
127	Thermometer, Kinematic Range 97.5 to 102.5 degrees F.	12
137	Thermometer, Dble. Diamond Brand, Engraved —30 to 0 degrees F	6
138	Thermometer, Dble. Diamond Brand, Engraved —61 to 29 degrees F	6
139	Thermometer, Dble. Diamond Brand, Engraved —90 to 60 degrees F	6
175	Wash Bottle, 500 ml cap	6
280	Flask, 500 ml Round Bottle Short Ring Neck	6
177	Bottle for Oil Sampling, 4 oz.	12
171	Bottles	13
278	Flask—1,000 ml	3
164	Beaker—1,000 ml	6
265	Graduated Cylinder 500 ml	3
452	Flexible Metal Braided Hose, ½" I.P.S.	3
453	Flexible Metal Braided Hose 1" I.P.S.	1
66	Thermo Hydrometer, large, 0–11 degrees	2
67	Thermo Hydrometer, large, 10–21 degrees	2
68	Thermo Hydrometer, large 19–31 degrees	4
69	Thermo Hydrometer, large, 29–41 degrees	4
70	Thermo Hydrometer, large, 39–51 degrees	4
71	Thermo Hydrometer, large, 49–61 degrees	8
72	Thermo Hydrometer, large, 59–71 degrees	4
73	Thermo Hydrometer, large, 69–81 degrees	2
74	Thermo Hydrometer, large, 79–91 degrees	2
122	Thermo Range —30 to 120 degrees F. (Vap. Press. App.)	6
339	Thermo, Gen. Purpose, —5 to 300 degrees F.	6
340	Thermo, Gen. Purpose —20 to 150 degrees C.	6
25	Thermo, —112 F. to 70 degrees F.	12
22	Thermo, —36 degrees F. to 120 degrees F.	6
129	Thermometer Kinematic, Range 207.5 to 212.5 degrees F.	12
128	Thermometer Kinematic, Range 127.5 to 132.5 degrees F.	12
121	Thermometer, 94–108 degrees F.	6
75	Thermo Hydrometer 4 oz. size 5–11 degrees	2
76	Thermo Hydrometer 4 oz. size 10–16 degrees	2
77	Thermo Hydrometer 4 oz. size 15–21 degrees	4
78	Thermo Hydrometer 4 oz. size 20–26 degrees	4
79	Thermo Hydrometer 4 oz. size 25–31 degrees	4
80	Thermo Hydrometer 4 oz. size 30–36 degrees	4
81	Thermo Hydrometer 4 oz. size 35–41 degrees	4

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TABLE VI. *Packing List*—Continued

Item No.	Description	Quantity
82	Thermo Hydrometer 4 oz. size 40–46 degrees	4
83	Thermo Hydrometer 4 oz. size 45–51 degrees	4
84	Thermo Hydrometer 4 oz. size 50–56 degrees	4
85	Thermo Hydrometer 4 oz. size 55–61 degrees	8
86	Thermo Hydrometer 4 oz. size 60–66 degrees	4
87	Thermo Hydrometer 4 oz. size 65–71 degrees	2
88	Thermo Hydrometer 4 oz. size 70–76 degrees	2
	Unit No. 10: One Armorply Shipping Crate and Platform. One Enameled Steel Cabinet, w/s.s. top. One Sliding Pipe Rack, Complete.	
171	Bottle, Narrow Mouth, Clear Round 1 Qt. Cap. Cork Top	13
181	Bottle, Reagent, w/glass stoppers, clear glass 16 oz.	36
350	Triangle, Chromel Wire, Silica No. A Ref: Cenco 19715	6
351	Triangle, Chromel Wire, Silica No. B Ref: Cenco 19715	6
352	Triangle, Chromel Wire, Silica No. C Ref: Cenco 19715	6
167	Beaker, 800 ml Ref: Fisher 2–540	12
164	Beaker, 1000 ml Ref: MS 35992–9	6
452	Flexible Metal Braided Hose, ½" I.P.S.	3
453	Flexible Metal Braided Hose, 1" I.P.S.	1
98B	Tubes, Test 50 ml Steam Emulsion App. Component Part	2
144	Wire Gauze, 16 Mesh, 6" x 6", Asbestos Center	6
98C	Test Tube 22 mm x 200 mm	2
302	Paper, Filter #12	4
	Unit No. 11: One Armorply Shipping Crate and Platform One Enameled Steel Cabinet w/s.s. top. One Sliding Pipe Rack, complete.	
263	Cylinder, Pyrex Glass, Lifetime Red, 100 ml	12
162	Beaker, 400 ml	24
161	Beaker, 250 ml	6
165	Beaker, 30 ml	6
273	Flask, Erlenmeyer, 50 ml	6
274	Flask, Erlenmeyer, 125 ml	6
163	Beaker, 600 ml	24
276	Flask, Erlenmeyer, 300 ml	12
281	Flask, Volumetric Cap. 1 Liter	6
452	Flexible Metal Braided Hose ½" I.P.S.	3
453	Flexible Metal Braided Hose 1" I.P.S.	1
158	Beaker, 20 ml	7
159	Beaker, 50 ml	6
177	Bottle for sampling oil	12
	Unit No. 12: One Armorply Shipping Crate and Platform One Enameled Steel Cabinet w/s.s. top. One Sliding Pipe Rack, complete.	
263	Cylinder, Grad. 100 ml	12
267	Cylinder, Grad. w/standard tapered stopper 100 ml	6
268	Cylinder, Grad. w/standard tapered stopper 250 ml	3

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TABLE VI. *Packing List*—Continued

Item No.	Description	Quantity
275	Flask, Erlenmeyer, 200 ml	6
277	Flask, Erlenmeyer, 500 ml	12
385	Goggles, Safety Chemical	4
264	Cylinder, Lifetime Red Grad. 250 ml	6
266	Cylinder, Lifetime Red. Grad. 1,000 ml	2
262	Cylinder, Lifetime Red. Grad. 50 ml	6
279	Flask, Iodine 250 ml	6
283	Funnel, Separatory 250 ml	6
284	Funnel, Separatory 500 ml	2
452	Flexible Metal Braided Hose ½" I.P.S.	2
453	Flexible Metal Braided Hose 1" I.P.S.	1
	Unit No. 13: One Armorply Shipping Container and Platform. One Enameled Steel Cabinet w/s.s. top. One Sliding Pipe Rack, complete.	
444A	Flow Meter for Foam Tester Component part.	1
2	U-Tube for Aniline Point.	3
260	Cylinder, Lifetime Red, 5 ml.	6
444B	Cylinder, 1,000 ml for Foam Tester Component part.	1
444C	Diffusion Rod for Foam Tester Component part.	1
444D	Therm. Reg. for Foam Tester Component part.	1
341	Thermo-Reg Straight Form, Mercury to Platinum Contact	2
353	Tube, Test, Chemical w/rims—18 x 150–mm	12
200	Alcohol Burner	2
249	Crucibles, coors, size #1 30 ml	12
250	Crucibles, coors, size #2 50 ml	6
254	Crucibles, coors, size #1 30 ml (High Form)	12
252	Crucibles, coors, size #3	12
146	Condenser, Pyrex 400 mm	4
145	Water Determination Apparatus	2
271	Diamond Pencil	1
189	Brushes	3
255	Cutter, Glass, Griffin, type	6
297	Labels, Gum, Red Border, Size No. 201	6
305	Pencils, Red Wax	12
328	Steel Strips—Pkg	1
301	Paper, Filter Whatman #40, 110 mm dia	2
149	Receiving trap, Crank Case Dilution	6
147	Receiving trap, Water by Dist	6
207	Cement, Dekhotinsky—Medium 1 oz. Pkg.	6
208	Cement, Dekhotinsky—Hard 1 oz. Pkg.	6
209	Cement, Duco 1 oz. tubes	6
321	Splashgon	3
246	Cord, Asbestos, Dia. 1/16"—lb	2
312	Rings, Suberite, Cork Inside Dia. 90 mm Outside Dia. 140 mm	3
603	Alkacid Tester, Fisher PH	5
251	Crucibles, Coors Size #4, 140 ml	6
115D	Shield Component part.	1
400	Meter, Univ. AC–DC	1
213	Clamp, Beaker, Rubber Lined	6
120	Gauge, Vapor Pressure Range 0–45 lb	1
119	Gauge Vapor Pressure Range 0–15 lb	4
156	Barometer, Aneroid	1
319	Rule, Slide, Ortho Phase Log-Log Model #500 metal	2
347	Crucible Tongs 18"	1
348	Tongs, Beaker 12"	2
65A	Oxy. Gauge, Parr Bomb Component part.	1

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TABLE VI. *Packing List*—Continued

Item No.	Description	Size	Quantity
	Unit No. 14: One Armorply Shipping Container and Platform. One Enameled Steel Cabinet w/s.s. top. One Sliding Pipe Rack, complete.		
284	Funnel, Separatory, 500 ml		4
52	Beaker, Pyrex Brezeluis, 100 ml		24
17	Tube, Centrifuge, Grad. 100 ml		12
95	Flask, Sediment by Extraction		5
282	Flask, Filter Heavy Wall, 1,000 ml		6
148	Flask, 500 ml Round Bot. long neck		5
285	Funnel, Ribbed, Mooney Airvent 250 ml		6
286	Funnel, Analytical 75 mm dia		6
194	Bulb, Red, Rubber		6
100	Jar for Steam Emulsion		4
Item No.	**Description**	**Size**	**Quantity**
	Unit No. 15: One Armorply Shipping Container and Platform. One Enameled Steel Cabinet w/s.s. top. One Sliding Pipe Rack, complete.		
668	Sodium Sulfate, Anhydrous, ACS, MW 142.05	2	1 lb.
641	Methyl Orange (Indicator) CP	2	¼ lb.
608	Ammonium Molybdate MW 123595	1	1 lb.
611	Ammonium Sulfate, ACS	1	1 lb.
609	Ammonium Nitrate, ACS	1	¼ lb.
658	Potassium Iodide, Granular, ACS, MW 166.02	1	1 lb.
649	Phenolphthalein	2	¼ lb.
607	Ammonium Iodide, N.F.	1	¼ lb.
634	Iodine Resublime, ASC	1	1 lb.
637	Lead Oxide Mono (Litharge) PBO	1	1 lb.
667	Sodium Peroxide, ACS, MW 77.99	1	1 lb.
636	Lead Acetate, Normal, MW 379.35	1	1 lb.
625	Cupric Sulfate Powder	1	1 lb.
674	Tannic Acid, NW 322.22	1	1 lb.
615	Barium Chloride, Anhydrous, MW 208.27	1	1 lb.
618	Benzoic Acid—Gr. NBS Certificate	60	grams
635	Ferrous Sulfide Granular	1	1 lb.
669	Sodium Thiosulfate ACS, MW 248.19	1	1 lb.
665	Sodium Nitrate Crystal ACS, MW 85.01	1	1 lb.
652	Potassium Bromide, Crystal	1	¼ lb.
631	Hydrofluoric Acid 48%	1	¼ lb.
616	Barium Hydroxide	1	¼ lb.
659	Potassium Nitrate	1	¼ lb.
676	Zinc Oxide	1	¼ lb.
642	Methyl Red (Indicator) CP	1	¼ lb.
661	Silver Nitrate Crystal	2	¼ lb.
660	Potassium Permanganate	1	¼ lb.
638	Mercury Metal Sp. Gr. 13.546	1	5 lb.
648	Para-Nitro-Phenol Indicator	100	gms.
656	Potassium Ferrocynaide, Crystal, ACS, MW 422.39	1	1 lb.
628	Ferric Chloride	1	¼ lb.
653	Potassium Chloride, ACS, MW 74.55	1	1 lb.
613	Aniline, ACS, MW 93.12	3	1 lb.
624	Cobaltous Chloride Crystal	1	¼ lb.
619	Bromine, ACS, At. W. 79.92 Sp. Gr. 3.119	1	¼ lb.
670	Starch Corn Powder, USP	2	1 lb.
610	Ammonium Oxalate, ACS	1	¼ lb.

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TABLE VI. *Packing List*—Continued

Item No.	Description	Quantity	Size
675	Tripoli Powder	1	5 lb.
605	Ammonium Chloride, ACS, MW 53.50	1	1 lb.
622	Celite (Analytical Filter Aid)	1	1 lb.
666	Sodium Oxalate, MW 134.01	1	1 lb.
647	Oleic Acid, Tech	1	1 lb.
604	Ammonium Acetate ACS, MW 77.08	1	1 lb.
672	Sulfur (Surfur Flower)	1	1 lb.
633	8—Hydroxyguinoline (Ortho)	1	¼ lb.
651	Phosphorus Pentoxide, Anhydride, MW 141.96	1	1 lb.
629	Glycerin, White, USP	1	1 lb.
623	Chloroform, USP	1	1 lb.
601	Acetic Acid, Glacial, 99.5% ACS	2	1 lb.
663	Sodium Chloride White Tech	2	5 lb.
662	Sodium Carbonate, Anhydrous, MW 106.00	2	5 lb.
614	Asbestos Fibre, Acid Washed, For Gooch Crucible	1	1 lb.
654	Potassium Chromate, ACS MW 194.20	1	5 lb.
664	Sodium Hydroxide, Pellets, ACS, MW 40.01	2	5 lb.
657	Potassium Hydroxide, Pellets, ACS, MW 56.10	1	5 lb.
655	Potassium Dichromate Crystal, ACS, MW 294.21	2	5 lb.
65B	Bottle of Pellets (Parr Bomb)	1	Btls.
40A	Beaker—400 ml Component part	1	
148	Flask, Pyrex—500 ml. Round bottom, long neck	1	
40B	Burner, Dropping Point App Component part	2	
40C	Stirrer and Motor Dropping Point App. Component part	1	
181	Bottle, sample reagent w/glass stoppers, clear glass 16 oz. round	36	
40D	Therm. Holder Dropping Point App. Ref: Component part	1	
5A	Refractory Block (Carbon Residue) Ref: Component part	2	
13	Refractory Block (Carbon Residue)	1	
63	Tubing, flexible, metal 1/16" I.D.	4	
452	Flexible Metal Braided Hose ½" I.P.S.	3	
98A	Rubber Stopper, Steam Emulsion App. Ref: Component part	4	
175	Glassware—Sets Bottle washing 500 ml. cap w/top	6	

Item No.	Description	Quantity
	Unit No. 16: One Armorply Shipping Crate and Platform. One Enameled Steel Cabinet. One Sliding Pipe Rack, Complete.	
152	Weight, Balance, 1 mg—100 gr. Analytical, Complete	2
154	Balance Torsion	1
151	Balance Analytical, Voland #640	1
153A	Plate holder for Balances Component part	4
153B	Large Balance plates Component part	2
155	Weight Balance, 1 gr.—500 gr.—set	1
444	Foam Tester complete w/heating coil, stand and rod	1
444E	Relay Control Box for Foamtester Component part	1
452	Flexible Metal Braided hose ½" I.P.S.	1
453	Flexible Metal Braided hose 1" I.P.S.	1
153	Balance Analytical, Voland #220	1
444F	Stirrer, foam test*	1
	Beam, balance analy*	1

*Component parts.

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TABLE VI. *Packing List*—Continued

Item No.	Description	Quantity
	Beam, balance student*	1
	Rider, bal. 5 mg.*	1
	Rider, bal. 10 mg.*	1
	Knob, bal*	1
	Support, pan. torsion*	2
	Saddle, bal. analy.*	2
	Saddle, bal. student*	2
	Chain, foam test*	1
	Unit No. 17:	
	One Armorply Shipping Crate and Platform.	
	One Enameled Steel Cabinet.	
	One Sliding Pipe Rack, Complete.	
433	Sink—Dwg. No. 5-14-333	1
18	Chromometer, Saybolt, Complete	1
26	Colorimeter Tag—Robinson, Complete	1
46	Flash Point Pensky Martens Closed Cup, Complete	1
49	Flash Point Tag, Closed Tester	1
290	Neoprene, Rubber Gloves—pr Size 9	6
322	Sponge Squares 5¼" x 3½" x 1⅝"	12
343	Thief, Drum, Oil Resinex	5
360	Tubing, Rubber ¼" Bore, thin wall—ft	50
362	Tubing Rubber 3/16" Bore, thick wall—ft	50
363	Tubing Rubber ¼" Bore, thick wall—ft	50
364	Tubing Rubber 5/16" Bore, thick wall—ft	25
423	Rubber Aprons, Length 46", width 36"	6
223	Cloth, Emery No. 00—shts	24
224	Absorbent Cotton 12" wide 1 lb. (pkg)	3
289	Asbestos Glove—pr	2
	Circuit Breakers, Component Part of Electrical System (For Storage only)	3
452	Flexible Metal Braided Hose ½" I.P.S.	1
453	Flexible Metal Braided Hose 1" I.P.S.	1
375	Wool Steel #1	3
376	Wool Steel No. 0, Medium	3
377	Wool Glass Filtering	2
211	Chamois, 2 ft. sq. (for filtering gasoline) size B	3
218	Clamps, Burette, Adjustable, Castalloy	12
216	Clamps, Ex. Castalloy	12
214	Clamp, Double, Burette, Holder	4
378	Circular dial microscope	1
	Unit No. 18:	
	One Armorply Shipping Crate and Platform.	
	One Enameled Steel Cabinet.	
	One Sliding Pipe Rack, Complete.	
54	Clamp, Bench, Gum Stability App.	1
55	Liner, Gum Stability App.	24
56	Recording Pressure Gauge 12" range 0–200 lbs. in 2 lb. divisions, 2 pen types	2
53	Oxidation Stability App. Gas (2 hole bath) w/all accessories, 115 V. AC.	1
59	Washer, Neoprene	24
60	Bomb, Complete, Gum Stability App.	2
61	Recording Chart	2
62	Ink Recording, 2 oz. bottle red (3 ea. red and 3 ea. blue)	6
64	Wrench, Bomb, Gum Stability Apparatus	2

*Component parts.

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TABLE VI. *Packing List*—Continued

Item No.	Description	Quantity
57	Hose, Rubber, Air With check valve 6' lengths	1
171	Bottle, Narrow Mouth, Clear, round 1 Qt. cap Cork Top	8
452	Flexible Metal Braided Hose ½" I.P.S.	1
303	Filter Paper, Heavy white creped, 33 cm.	2
214	Clamp, Double, Burette, holder	4
215	Clamp, holder, castalloy	2
217	Clamp, Versatile, Castalloy	6
	Unit No. 19: One Armorply Shipping Crate and Platform. One Enameled Fume Hood w/I–H–Blower. Interior Lined w/Transite.	
318	Sheet Copper, 12" Wide by 60" long 24 Ga.	1
	Unit No. 20: One Armorply Shipping Crate and Platform. One Enameled Steel Cabinet w/s.s top. One Sliding Pipe Rack, Complete. One Elevating Platform. One Stainless Steel Cover.	
434	Kinematic Viscosimeter Complete with refrigeration equipment.	1
	Unit No. 21: One Armorply Shipping Crate and Platform. One Enameled Steel Cabinet w/s.s. top. One Sliding Pipe Rack, Complete.	
437	Refrigerator complete with refrigeration equipment Ref: Lawler Elec Co. Bayonne, N. J. (See Dwg 5–14–337)	1
	Unit No. 22: One Armorply Shipping Crate and Platform. One Enameled Steel Cabinet w/s.s. top. One Sliding Pipe Rack, Complete.	
438	Cloud and Pour Apparatus complete with refrigeration equipment	1
	Unit No. 23: One Armorply Shipping Crate and Platform.	
435	One compressor, Air, Ingersoll Rand, two stage unit, type 30, 200 lbs. per square inch. Continuous service 1 HP, tank size 20" x 50", 60 gallons, 60 cycle, 110–220 V, Single phase AC mounted to one starting switch.	1
436	Switch, motor protection for Air Compressor single phase, 110 volts Component part Ref: Renick & Mahoney, Inc. Catalog #232C–1	1
	Unit No. 24: One Armorply Shipping Crate and Platform. One Enameled Steel Sabinet w/s.s. top. One Sliding Pipe Rack, Complete.	
38	Flask, Distilling, 100 cc. Dist. App	10
39	Flask, Distilling, 250 cc. Dist. App	4
195	Burette, 50 ml cap., w/stopcock, blueline "EXAX" Retested	6
306	Pipette, Transfer 50 ml	6
307	Pipette, Transfer 25 ml	6
308	Pipette, Transfer 10 ml	6
12	Beads, Glass, 5 mm Dia.—pkg.	1
166	Beaker, 5 ml	6

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TABLE VI. *Packing List*—Continued

Item No.	Description	Quantity
41	Grease Cup Assembly	3
160	Beaker, Griffin 100 ml cap	6
97	Thimble, Alundum, Sediment by Extraction	12
93A	Flask, Narrow Mouth-Sed. by Extraction 1,000 ml Complete	2
94	Condenser, Metal Extraction	2
23	Jar, Pour test (2 colorimeter)	12
253	Crucible Holder, Walters, for size No. 3 Gooch Crucible	2
329	Stopcock, Glass, Pyrex, Two-way, Straight Bore 2 mm. Ref: Cenco 15406	6
452	Flexible Metal Braided Hose ½" I.P.S.	1
453	Flexible Metal Braided Hose 1" I.P.S.	2
1A	U-Tube Stirrer for Aniline Pt. App. (component)	1
291	Grease, Stopcock, 1 oz. Tube	6
15	Cup, Shield Alum. 100 ml centrifuge	4
96	Support, Thimble, Sed. by Ext	6
	Unit No. 25: One Armorply Shipping Crate and Platform. One Enameled Steel Cabinet w/s.s. Top. One Sliding Pipe Rack, Complete.	
14	Centrifuge, Electric, Complete	1
439	Channel Apparatus complete w/centrifuge equipment	1
24	Cloud and Pour Apparatus, 4 Unit, All Metal	1
	Unit No. 26: One Armorply Shipping Crate and Platform. One Enameled Steel Cabinet, w/s.s. top. One Sliding Pipe Rack, complete.	
168	Borer, Cork	1
325	Stretcher, Rubber tubing	6
243	Sharpener, Cork Borer	1
219	Clamp, Castalloy, Hosecock	24
345	Tongs, Crucible 9" Chrome	4
220	Clamp, Pinchock, Brass Flat Jaws 2¾"	12
316	Shears, 6" Gen. Utility	3
324	Spatula 6" Blade	6
315	Scissors, Dissecting, Med. Weight, Cutting Edge 40 mm, Overall Length 115 mm	2
346	Tongs, Pick-up 8"	2
369	Watch, Glasses—35 mm	6
370	Watch, Glasses—65 mm	6
21	Gasket Saybolt Chromometer G17585	12
37	Rubber, Washer Dist. App.	12
247	Cots, Finger, Style A	4
248	Cots, Finger, Style B	2
261	Cylinder, Grad. 10 mm	6
136	Viscosimeter, Cannon Fenske Opaque No. 600	4
130	Pipettes, Kinematic, Fenske Ostwald ASTM, Calibrated No. 50	2
132	Pipette, Kinematic Fenske Ostwald ASTM, Calibrated No. 200	2
133	Pipette, Kinematic Fenske Ostwald ASTM, Calibrated No. 300	2
134	Pipette, Kinematic Fenske Ostwald ASTM, Calibrated No. 400	2
135	Pipette, Kinematic Fenske Ostwald ASTM, Calibrated No. 500	2
323	Spatula 4" Blade	6
349	Forceps, Monel Metal 6" Gooseneck	2
98	Steam Emulsion App. Complete Glassware 10 pcs	2
42	Test Tube for Dropping Pt. Apps	6
173	Bottle Babcock Unsaturation test Standard Solvent	6
371	Watch Glass 100 mm	12

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TABLE VI. *Packing List*—Continued

Item No.	Description	Quantity
19	Tube, Glass, Plain Chromometer	2
20	Tube, Glass, Grad. Chromometer w/connection	2
309	Policeman	12
99	Tubes, Oil Emulsion	6
296	Cylinders (Hydrometer Jars) 50 x 350 mm	6
445	Sampling Devices, ASTM 1 Qt. ¾" openings	4
1	Aniline Pt. App. S.I.L. Stirrer and Jar	1
5	Carbon Residue Apparatus Cover	2
6	Outer Crucible and Cover, Nickel Carbon Res. Cover	1
8	Crucible Skidmore w/nickel cover Carb. Res Cover	1
9	Hood, Nickel Car. Res. App	1
10	Chromel Triangle, Car. Res. App	8
11	Burner, Meeker type, Car. Res. App	1
245	Cork Screw	2
27	Colorimeter, ASTMD 1500, (minus 2 jars)	1
452	Flexible Metal Braided Hose ½" I.P.S.	3
453	Flexible Metal Braided Hose 1" I.P.S.	1
327	Component part Stirrer, Elect	2
	Unit No. 27: One Armorply Shipping Crate and Platform.	
115	Reid vapor pressure apparatus	1
	Unit No. 28: One Armorply Shipping Crate and Platform. One Enameled Fume Hood w/1–H Blower Interior Lined w/transite	
204	Can. Screw top, 5 gal. capacity	6
221	Cloth, Cheese, Single Weights	100 yds
	Case No. 29/1	
354	Tubing, Glass Pyrex OD 6 mm	10 lbs.
355	Tubing, Glass, Pyrex OD 8 mm	10 lbs.
356	Tubing, Glass Pyrex, OD 10 mm	10 lbs.
357	Tubing, Glass Pyrex, OD 12 mm	10 lbs.
140	Rod, Glass Pyrex, OD 6 mm, 4 ft. length	5 lbs.
141	Rod, Glass Pyrex, OD 8 mm, 4 ft. length	5 lbs.
142	Rod, Glass Pyrex, OD 9 mm, 4 ft. length	5 lbs.
143	Rod, Glass Pyrex, OD 10 mm, 4 ft. length	5 lbs.
	Case No. 29/2	
	Corks, regular length	
225	No. 2	100
226	No. 3	100
227	No. 4	100
228	No. 5	100
229	No. 6	100
236	No. 14	100
237	No. 15	100
238	No. 18	100
241	No. 24	100
242	No. 26	100
	Case No. 29/3	
272	Flexframe, Set #1 Comp	2
98 & 145	Rods, Metal, for App., Steam Emulsion (2 ea.) Component parts	4
183	Brush, Beaker, Large	6
187	Brush, Burette, Small	3

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TABLE VI. *Packing List*—Continued

Item No.	Description	Quantity
188	Brush, Pipette	3
18	Brush, Flask and Bottle	6
184	Brushes, Beaker Small	3
185	Brushes, Large, Test Tube	3
186	Brushes, Small, Test Tube	3
191	Brush, Wire	1
193	Brushes, Cylinder, Large	3
332	Rods, Long Support Component part.	6
331	Rods, Short Support Component part.	1
	Case No. 29/4	
295	Hot Plate, 8" diam., 100 v., AC	2
	Case No. 29/5	
170	Bottle, Narrow Mouth, Clear Bakelite, Screw Cap. French Square 1 Qt. capacity	24
	Case No. 29/6	
171	Bottles, Narrow Mouth, Bakelite, Screw Cap, Amber French Square 32 oz.	12
29	Board, Asbestos 4" Hole, 8" x 11" x ¼"	2
30	Board, Asbestos, 2¾" Hole, 6" x 6" x ¼"	3
31	Board, Asbestos, 1½" Hole, 6" x 6" x ¼"	4
32	Board, Asbestos, 1¼" Hole, 6" x 6" x ¼"	4
293	Block, Top, Refractory, upper heater Porcelain, 3⅛" Hole	4
	Case No. 29/7	
206	Cans, Tin, 1 Qt. Capacity w/Screw neck and Cap With Innerseals (for lubricating oils)	34
	Case No. 29/8	
203	Can, Screw Top, 1 gal. Capacity	12
	Case No. 29/9	
203	Cans, Screw Top, 1 Gal. Capacity	6
206	Cans, Tin, 1 Qt. Capacity With Screw Neck and Cap With Innerseal (for lubricating oils)	18
	Case No. 29/10	
176	Bottle, Clear, Screw Cap, Rd Narrow Mouth 2 oz.	12
179	Bottle, Reagent, Narrow Mouth Clear, Glass 2 oz.	12
	Case No. 30/11	
65	Sulfur Bomb, Petroleum Oils, Parr. Elec. Ignition ASTM D–129–39 Complete	1
	Case No. 30/12	
180	Bottle, Reagent Drop CP Narrow Mouth, Brown Hood Stopper, 60 ml	36
	Case No. 30/13	
169	Bottle, Clear Glass, Screw Cap, Narrow Mouth, French Square, 8 oz.	24

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TABLE VI. *Packing List*—Continued

Item No.	Description	Quantity
	Case No. 30/14	
172	Bottle, Narrow Mouth, Bakelite, Screw Cap, Amber French Square, 30 oz.	24
	Case No. 30/15	
172	Bottle, Narrow Mouth, Bakelite, Screw Cap, Amber French Square, 30 oz.	24
	Case No. 30/16	
206	Can, Tin, 1 Qt. Cap With Screw Neck and Cap With Innerseals	72
	Case No. 30/17	
203	Can, Tin, 1 Gal. Cap With Screw Neck and Cap With Innerseals	18
	Case No. 30/18	
206	Can, Tin, 1 Qt. Cap With Screw Neck and Cap With Innerseal	36
	Case No. 30/19	
203	Can, Screw Top, 1 Gal. Capacity	12
	Case No. 30/20	
113	Tetraethyl Lead, Extraction Apparatus (Stands and Rods)	3
	⅜" x 20" Long Rod Component Part	6
	Case No. 31/21	
126	Standardization Oil, 100 F. 122 F. & 210 F.	4
	Case No. 31/22	
292	Heaters, Elect. Full Kontrol 740 W.	4
	Case No. 31/23	
332	Support Stand Rect. Base Large Size 6" x 9"	6
331	Support Stand Rect. Base Medium Size 5" x 8"	6
190	Brush, Sink	2
330	Stand, Tripod, w/Concentric Rings, 9" x 6"	4
333	Ring Support, Iron, w/Clamp, Size 5"	3
335	Ring, Support, Iron w/clamp, Size 3"	3
334	Ring, Support Iron w/clamp, Size 4"	3
	Support, Stand 5" x 8" Component part	4
	Stand, Tripod w/rods Component part	4
	Stand, Tripod Rings Only Component part	2
	Rack for Water Still Component part	1
	Case No. 31/24	
174	Bottle, Wide Mouth, Clear, Glass, Screw Cap 4 oz.	48
	Case No. 31/25	
44	Flash and Fire Pt. Cleveland Open Cup, Complete Set w/750 Watt Heater—115 V. 50–60 cycles, AC	1

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TABLE VI. *Packing List*—Continued

Item No.	Description	Quantity
	Case No. 31/26	
	Full-Kontrol Heaters	
40 & 145	550 W. (Item #145, 2 ea.)	3
4	750 W. Special For Aniline Point Apparatus	1
	Case No. 31/27	
205	Cans, Friction, Top, 2 lb. capacity (for Grease)	12
	Case No. 31/28	
269	Dessicator, 250-mm dia. w/porcelain plate	2
	Case No. 31/29	
269	Dessicator, 250-mm dia. w/porcelain plate	2
	Case No. 31/30	
	Corks Regular	
230	#7	100
231	#8	100
232	#9	100
233	#10	100
234	#11	100
235	#12	100
239	#20	100
240	#22	100
	Ref: Fisher 7–790.	
313	Stoppers, Rubber, 2 hole Assorted	10 lbs.
314	Stoppers, Solid, Rubber, Assorted	10 lbs.
	Case No. 31/31	
178	Bottle, Narrow Mouth, 5 Gal. Capacity Clear, Round	3
	Unit No. 33	
606	Ammonium Hydroxide Sp. Gr. 0.900	8 lbs.
643	Nitric Acid Sp. Gr. 1.42, ACS	21 lbs.
673	Sulfuric Acid Sp. Gr. 1.84, ACS MW 98.08	4–9 lb-btls.
	Unit No. 34	
620	Calcium Chloride, Calcine Porous, Tech 8 mesh	1–25 lb.
	Unit No. 35	
632	Hydrogen Peroxide Tech	10 lb.
621	Carbon Tetrachloride Tech	1–5 gal.
	Unit No. 36	
602	Acetone, U.S.P. Tech	1–5 gal.
639	Methyl Alcohol Absolute, ACS	1–1 gal.
612	Anhydrous Isopropyl Alcohol	1–1 gal.
	Unit No. 37	
617	Benzene (Benzol) ACS, Sp. Gr. 0.897	1–5 gal.
644	Naphtha Solvent ASTM D91–40	2–1 gal.
	Unit No. 38	
627	Ethylene Glycol Purified	1–5 gal.
645	Naphtha Solvent Xylene Tech N–2	2–1 gal.

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TABLE VI. *Packing List*—Continued

Item No.	Description	Quantity
	Unit No. 39	
645	Naphtha Solvent Xylene N–2	2–1 gal.
644	Naphtha Solvent ASTM D91–40	1–1 gal.
640	Methyl Ethyl Ketone, Tech	1–5 gal.
	Unit No. 40	
650	Petroleum Ether	1–5 gal.
644	Naphtha Solvent ASTM D91–40	2–1 gal.
	Unit No. 41	
646	Naphtha Solvent Xylene, D–95–46	3–1 gal.
671	Stoddard Solvent	1–5 gal.
	Unit No. 42	
646	Naphtha Solvent, ASTM D95–46	2 gal.
645	Naphtha Solvent, Xylene, Tech N–2	1 gal.
	Unit No. 43	
419	Iso-Octane w/blending table, 55 gallon drums	1
	Unit No. 44	
419	Iso-Octane w/blending tables 55 gallon drums	1
	Unit No. 45	
420	Normal Heptane w/blending tables, 55 gal. drum	1
	Unit No. 46	
420	Normal Heptane w/blending tables, 55 gal. drum	1
	Unit No. 47	
257	Cylinder, Oxygen Gas, 220–240 cu. ft. filled w/oxygen gas	1
	Unit No. 48	
257	Cylinder, Oxygen Gas, 220–240 cu. ft. filled w/oxygen Gas	1
	Unit No. 49	
256	Cylinder, Liquefied, Petroleum Propane (Filled) 100 lbs. with valve	1
	Unit No. 50	
256	Cylinder, Liquefied Petroleum Propane (Filled) 100 lbs. with valve	1
	Unit No. 51	
701	Model V16, Engine Fuel Testing, C. O. T. Military, Motor and Research Methods, Complete, for one base petroleum laboratory	1
	Unit No. 52	
701	Parts for C. O. T. Engine, Components parts	1
	Unit No. 53 (Parts for Waukesha Motor)	
743	Gauge Oil Temp	1
724	Ignition Coil	1
717	Volt Meter	1
742	Amp. Meter	1

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TABLE VI. *Packing List*—Continued

Item No.	Description	Quantity
	Carburetor Fuel Jets:	
729	50 cc	3
730	52 cc	3
731	54 cc	3
732	56 cc	3
733	58 cc	3
734	60 cc	3
735	62 cc	3
736	64 cc	3
737	66 cc	3
738	68 cc	3
739	70 cc	3
	Ref: Waukesha 75988–A	
740	Plug, Spark	1
706	Fittings ⅛" straight	1
718	Wrench Adj. End	1
722	Screw Driver	1
714	Wrench Crank Gear Nut	1
716	Wrench Flywheel Nut	1
715	Wrench Spark Plug	1
744	Wrench Handle	3
728	Cap. Carburetor Float Gauge	1
762	Assembly Cond. Gauge Level	1
705	Nut ⅛" Flared pressure line	1
748	Manifold Heater Housing, Flange Type	
	Ref: Waukesha #106747–A.	
702	Gasket, Carburetor and Exhaust Manifold	2
703	Tube, Neon, Spark Indicator	2
704	Washer, Carburetor Horizontal Jet Hole	2
707	Washer, Carburetor Float Gage Glass, Top	4
708	Washer, Carburetor Fuel Tank (Lower)	2
709	Washer, Condenser Gage, Lower	2
710	Washer, Carburetor Float Gage Glass, Bottom	2
711	Washer, Condenser Gage & Fuel Tank Gage (Upper)	2
712	Glass, Carburetor Fuel Tank Gage	2
713	Gasket, Water Pipe	2
723	Gasket, Carburetor Fuel Tank Gage	2
725	Gasket, Condenser Body	2
727	Glass, Carburetor Float Gage	2
741	Thermometer, Oil, Water, Air, Range 0–220 F.	1
745	Fuse, Plug, Panel	2
746	Gasket, Manifold	2
747	Thermometer, Air Intake, Range 60–160 F.	2
749	Thermometer, Heater, Range 300–400 F.	1
750	Valve, Intake, Shrouded	1
751	Valve, Exhaust	3
752	Ring, Piston, Compression (Tapered)	4
753	Ring, Piston, Oil (Square)	2
754	Ring, Piston, Oil (Tapered)	1
755	Burette	2
757	Gasket, Carburetor, Thick, 2 Bolt	2
758	Gasket, Water Inlet Cap, Coolant Drainpipe	1
759	Screwdriver, Allen	1
760	Pliers, Piston-Pin Retaining Ring	1
761	Fuse, Detonation Meter	1
769	Tachometer, Mechanical, Hand Held	

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TABLE VI. *Packing List*—Continued

Item No.	Description	Quantity
	Unit No. 54 (Tool Boxes) Kwick-Way Kits	
380	Compound, Grinding and Lapping Grease Mixed, 4 oz. Can Carb. #277 or Equal	6
383	Gauge, Cylinder, Dial type, Range 2½"–6" w/comb. Rigid and Toggle Joint Handle, Locking Mechanism	1
430	Valves, Globe, Brass ¼"	6
399	Diamond Dressing Tool, for Kwick-Way	1
386	Kwickway CFR Seal Grinder, 110 V., Motor with special small Bearing Grinder unit for operation on special 1 1/32" top arbors, wheel dressing unit and steel carrying case	1
431	Valves, Globe, Brass ⅜"	6
381	File, Rat tail 5"	2
382	File, 3-cornered, 6" tapered	2
384	Gauge, Thickness 26 leaves, range .0015 to .025	2
401	Oiler, ⅓ pt. Cap	1
402	Puller, 2 way Nail No. 1AVX3	1
403	Screw Driver, 3" Blade	2
404	Screw Driver, 6" Blade	2
405	Solder, Resin Center, 1 lb. spools	2
406	Solder Acid Core, 1 lb. spools	2
408	Wrench pipe adj. 6"	2
409	Wrench pipe adj. 8"	2
410	Wrench pipe adj. 10"	2
411	Wrench pipe adj. 14"	1
412	Wrench pipe adj. 18"	1
413	Wrench crescent adj. 6"	1
414	Wrench crescent adj. 8"	2
415	Wrench crescent adj. 10"	2
416	Wrench crescent adj. 12"	1
417	Wrench crescent adj. 15" Above wrenches standard type of equipment.	1
418	Iron, Soldering, Elect. ½ lb.	1
424	Union nuts, tube odd sizes ⅜ W41 x 6	6
425	Elbow union tube odd sizes ⅜ W55 x 6	6
426	Elbow male tube odd size ⅜ W49 x 6	6
427	½" union tube odd size ⅜ W48 x 6	6
428	3 Way "T" union odd size ⅜ W44 x 6	6
429	3 Way "T" union odd size ⅜ W51 x 6	6
317	Shears 12½" gen. utility	2
719	Wrench Box Socket 11/16 x ¾	1
720	Wrench Box Socket ½ x 9/16	1
721	Wrench Box Socket ⅜ x 7/16	1
726	Breaker Point Stone	1
756	Feeler Gauge	1
763	Cylinder Pipe	1
	Unit No. 55	
379	Calipers, Micrometer, Outside, w/lock nut and ratchet stop, comp. w/testing rods and case. Range 0"–6" x .001"	1
387	Kwickway CFR Seat Grinder (Spec. Eccentrimeter)	1
	Unit No. 56	
421	Benzene Nitration-Grade 55 gal. drum ASTM D835–47	1–55

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TABLE VI. *Packing List*—Continued

Item No.	Description	Quantity
	Unit No. 57	
630	Hydrochloride Acid-lbs. Sp. Gr. 1.19	30
	Unit No. 58	
258	Cylinder, Carbon Dioxide, CO_2 Gas, w/valve, 50 lbs. filled	1
	Unit No. 59	
258	Cylinder, Carbon Dioxide, CO_2 Gas, w/valve, 50 lbs. filled	1
	Unit No. 60	
259	Cylinder, Gas Freon, 12, 5 lb. Container Filled	1
	Unit No. 61	
422	Reference Fuel Grade Toluene Standardization With Calibration Table in 55 Gallon Drum	1 Drum
	Unit No. 62	
159	Beaker, Griffin 50 ml Cap	6 pcs
168	Borer, Cork	1 set
158	Beaker, Griffin 20 ml	5 pcs
186	Brush, Small, Test Tube	3 pcs
189	Brush, Balance	3 pcs
250	Crucible, Coor, Size #2, 50 ml	6 pcs
166	Beaker, Griffin, 5 ml Cap	6 pcs
271	Scriber, Pencil, Diamond Writing	1 pc
304	Pencil, Carborundum	2 pcs
320	Indicator, Speed, Improved Form	1 pc
368	Watch, Stop, 60 Second Dial	8 pcs
369	Watch, Glass, Concave, 35 mm	6 pcs
370	Watch, Glass, Concave, 65 mm	6 pcs
381	File, Rat Tail, 5"	1 pc
382	File, 3 cornered 6" tapered	1 pc
413	Wrench, crescent, adj. 6"	1 pc
90	Cone, Grease	2 pcs
50	Thermometer, Closed Tester, Range −4 to 122 deg. F.	6 pcs
114	Viscosimeter, Thermo, Saybolt	1 Set
161	Beaker, Griffin, 250 ml Cap	6 pcs
165	Beaker, Griffin, 30 ml	6 pcs
185	Brush, Large, Test Tube	3 pcs
188	Brush, Pipette	3 pcs
191	Brush, Wire	1 pc
192	Brush, Cylinder, Small	2 pc
193	Brush, Cylinder, Large	3 pc
194	Bulb, Red Rubber	6 pcs
196	Burner, Wing, Top, 11 mm OD	3 pcs
199	Burner, Pittsburgh, Univ	4 pcs
201	Burner, Micro	1 pc
266	Cylinder, Pyrex Glass, Lifetime Red 1,000 ml	4 pcs
273	Flask, Erlenmeyer, Pyrex Vial Mouth 50 ml	6 pcs
187	Brush, Burette, Small	3 pcs
388	Special Arbor w/$1\frac{1}{32}$" top, size $\frac{5}{16}$ CFR Grinder*	1 pc
389	Special Arbor w/$1\frac{1}{32}$" top, size $\frac{5}{16}$" Plus .005 Grinder*	1 pc
390	Special Arbor w/$1\frac{1}{32}$" top, size $\frac{5}{16}$" Plus .001 Grinder*	1 pc
391	Special Arbor w/$1\frac{1}{32}$" top, size $\frac{5}{16}$" plus .0015 Grinder*	1 pc
392	Special Arbor w/$1\frac{1}{32}$" top, size $\frac{5}{16}$" plus .002 Grinder*	1 pc

*Contractor to locate above items, in drawers and cabinets wherever possible.

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TABLE VI. *Packing List*—Continued

Item No.	Description	Quantity
393	Arbors, Size 7/16" plus .002 Grinder*	1 pc
394	All purpose Grinding Wheel 1⅜ x 45°*	6 pcs
395	All purpose Grinding Wheel 1 7/16 x 15°*	6 pcs
398	Flexible Rubber Hose Couplings	12 pcs
7	Cruc. Porcelain, Coor Size #1, Carb, Res. App.	12 pcs
64	Wrench, Bomb, Gum Stab. App.	1 pc
198	Burner Hi-Temp	10 pcs
253	Cruc. Holder, Walters, for #3 Gooch Cruc.	4 pcs
222	Timer, Victor, Interval, Manual	1 pc
264	Cylinder, Pyrex Glass, Lifetime, Red, 250 ml	6 pcs
265	Cylinder, Pyrex Glass, Lifetime, Red, 500 ml	3 pcs
268	Cylinder, Pyrex Glass, w/standard tapered stopper	3 pcs
270	Dish, Evaporating Coors, Porcelain, Size 0, 80 ml	6 pcs
276	Flask, Erlenmeyer, 300 ml	12 pcs
295	Hot Plate, 8" dia. 110 V., AC	1 pc
312	Rings, Suberite, Cork Inside 90 mm. Outside 140 mm	3 pcs
274	Flask, Erlenmeyer, 125 ml	6 pcs
195	Burette, 50 ml, capacity w/stopcock, blue line EXAX retested	6 pcs
639	Methyl Alcohol Absolute, ACS-gal.	4
310	Pump Vacuum, "Hyvac" w/iron base, Serial "A": w/motor 110 V., 60 cycle	1 pc
440	Regulator, Oxygen	1
441	Burette, Octane Blending	4
446	Strip, Copper, Greiner G–19236	12
447	Standards Corrosion ASTM Method 5313–Set	1 set
448	Bob, Plumb Brass, Outage 16 oz., Ref: Lufkin No. 587, Item used with item No. 336 in Unit No. 6	3
449	Bob, Plumb, Brass Innage 20 oz., Item used with item No. 336 in Unit No. 6	3
33	Cylinder, Grad. Glass 200 ml, 0–100% Special	6 pcs
63	Tubing, Flexible, Metal	4 pcs
160	Beaker, Griffin, 100 ml	6 pcs
202	Burner, Forbes, Bunsen	4 pcs
267	Cylinder, Pyrex, Grad. w/standard tapered stopper 100 ml	6 pcs
275	Flask, Erlenmeyer, 200 ml. Vial Mouth	6 pcs
278	Flask, Erlenmeyer, 1,000 ml. Vial Mouth	3 pcs
299	Mortar, Coors, Porcelain, w/pestle, size #1	1 set
150	Asbestos Fiber, Acid Wash, Short Fiber	1 lb.
342	Thermo-Regulator Guard	1 ea
197	Burner Gas, Fisher	4 pcs
288	System, Gas, Complete w/gauge and connections	1 set
326	Stopcock, Laboratory, Brass Inlet, ⅜" Male	24 pcs
212	Charts, Kinematic, Viscosity, Type "E" pads	2
210	Charts, Temperature, Viscosity, Index pads	2
626	Ethyl Alcohol, 95% USP—gal.	5
450	SOD Constant Temperature Bath	1 set
	Unit No. 63	
256	Cylinder, Liquefied, Petroleum Propane (Filled) 100 lbs. with valve	1
	Unit No. 64	
256	Cylinder, Liquefied, Petroleum Propane (Filled) 100 lbs. with valve	1

*Contractor to locate above items, in drawers and cabinets wherever possible.

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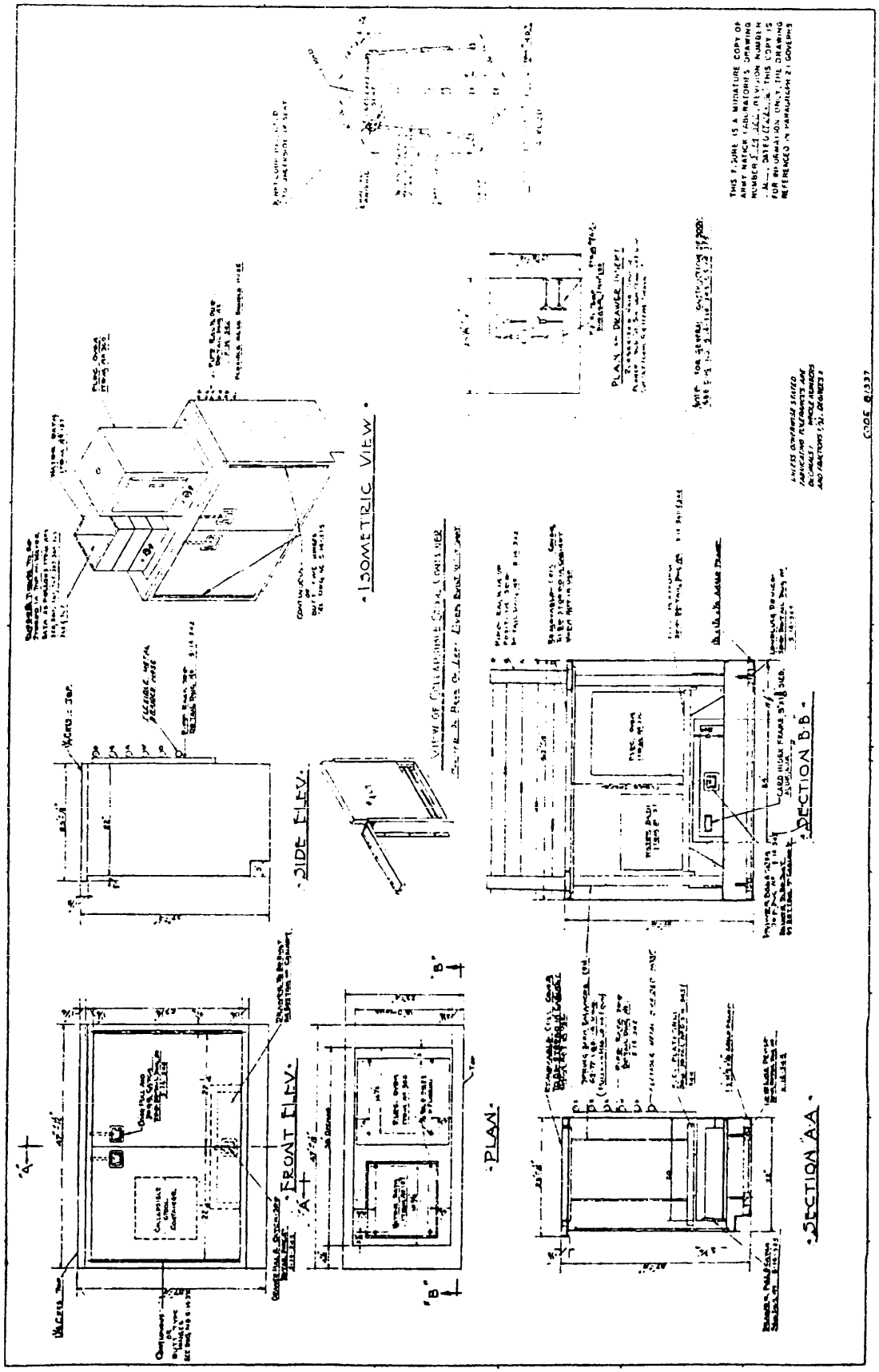


FIGURE 1. *Laboratory, petroleum, base; storage cabinet, unit No. 1, oven and water bath apparatus, assembly and details.*

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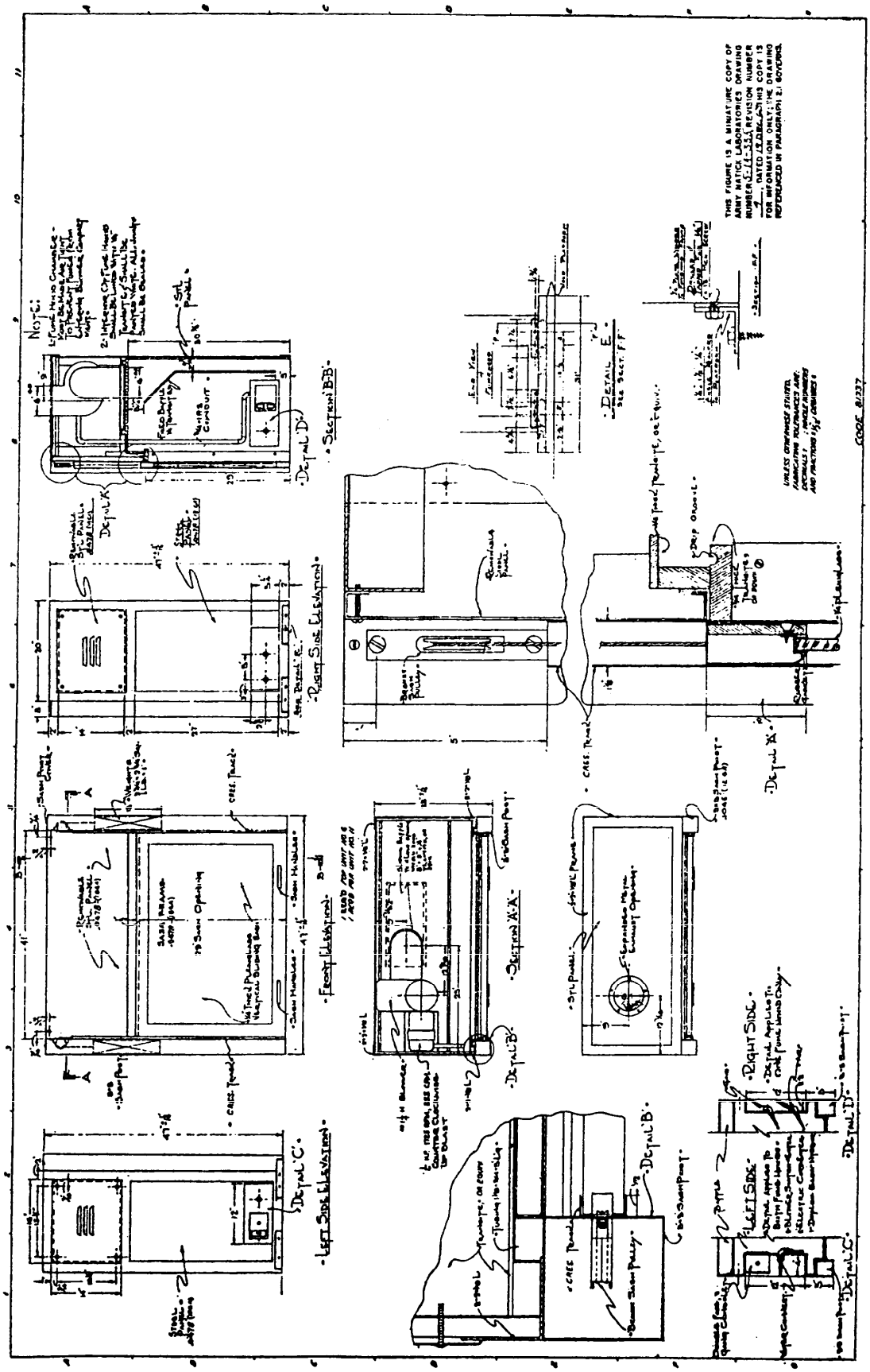


FIGURE 2. *Laboratory, petroleum, base; unit No. 19 fume hood.*

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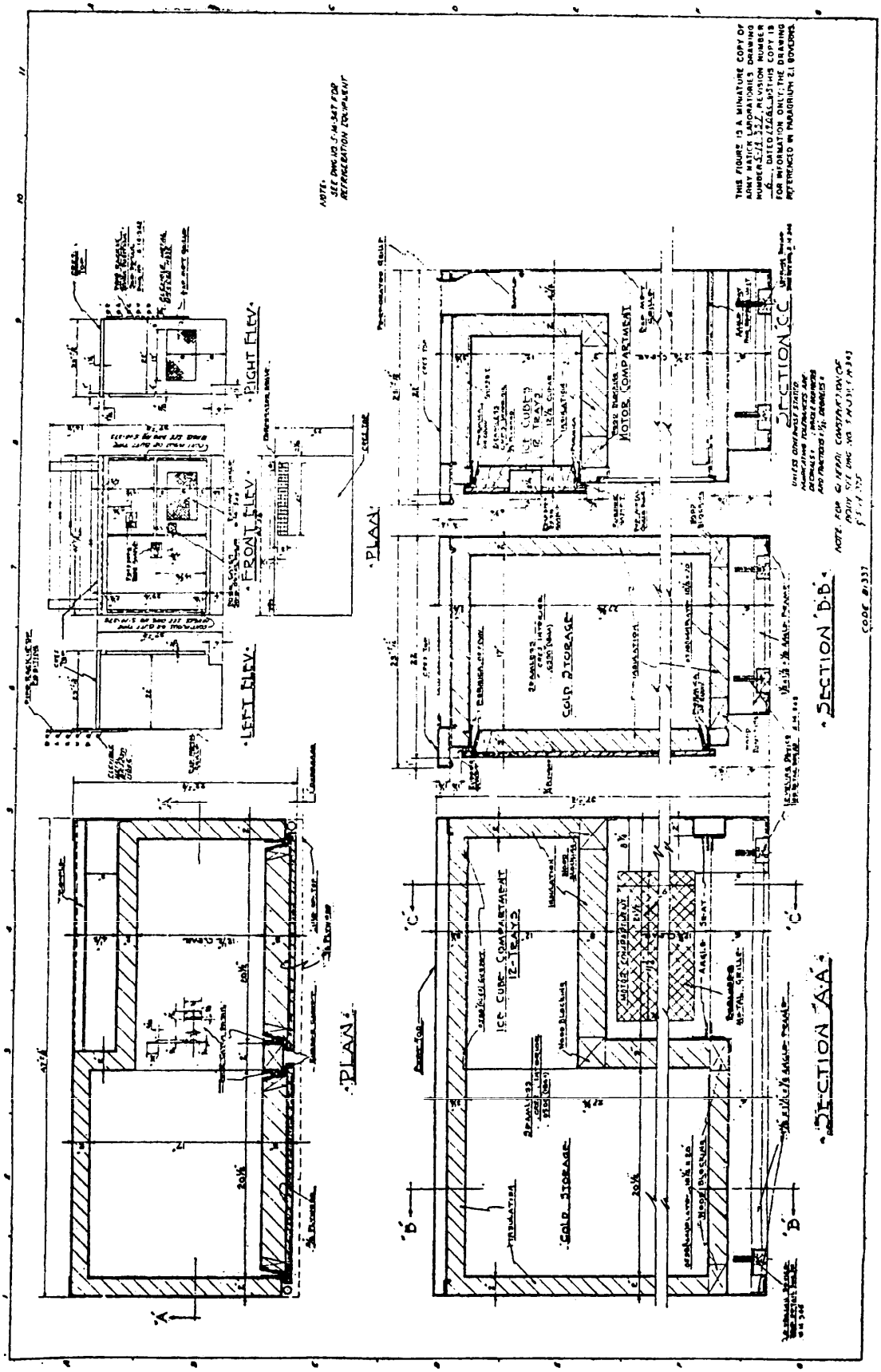


FIGURE 8. *Laboratory, petroleum, base; storage cabinet, unit No. 21, refrigerator cabinet, assembly and details.*

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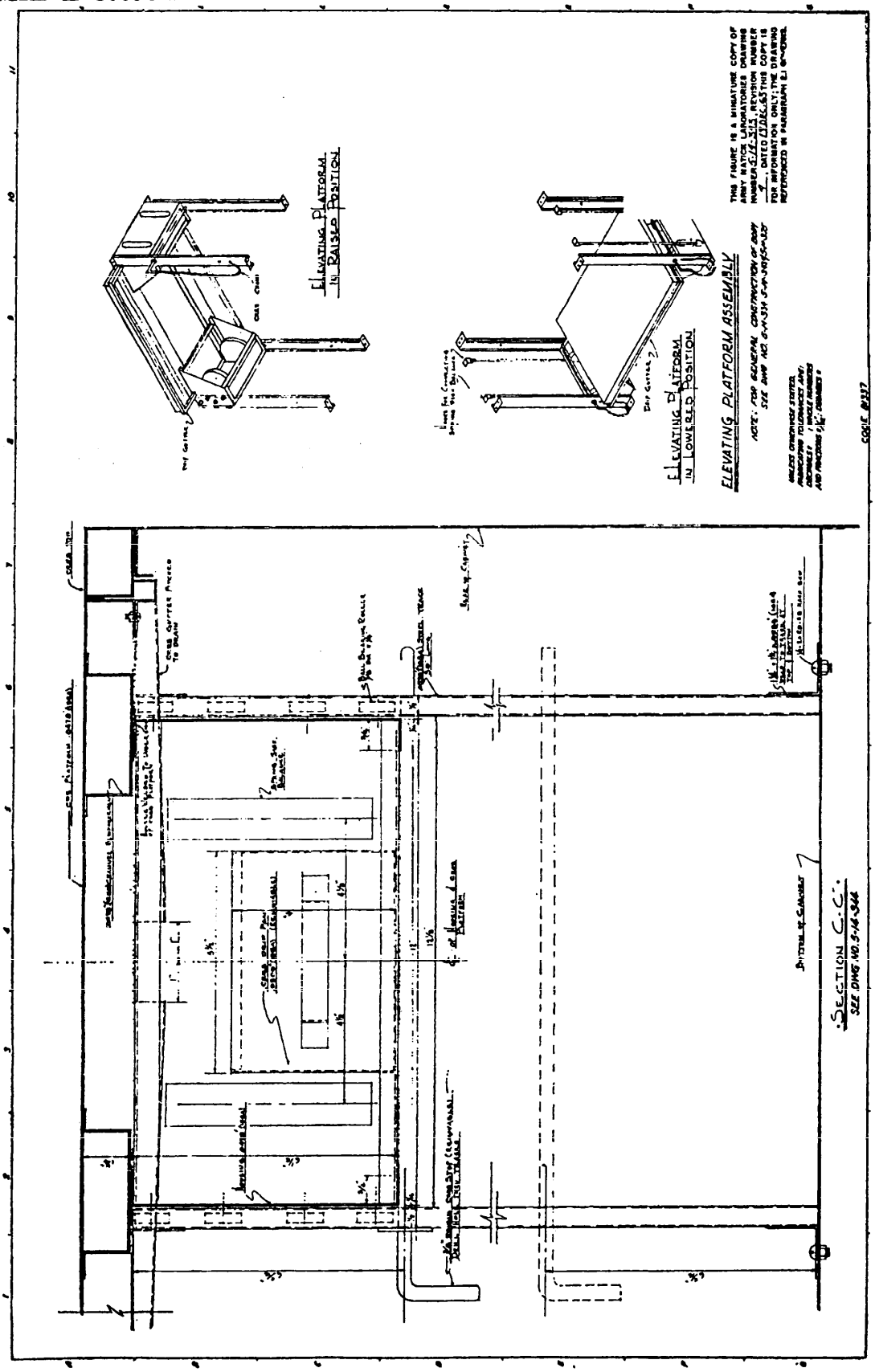


FIGURE 4. *Laboratory, petroleum, base; elevating platform mechanism, assembly and details.*

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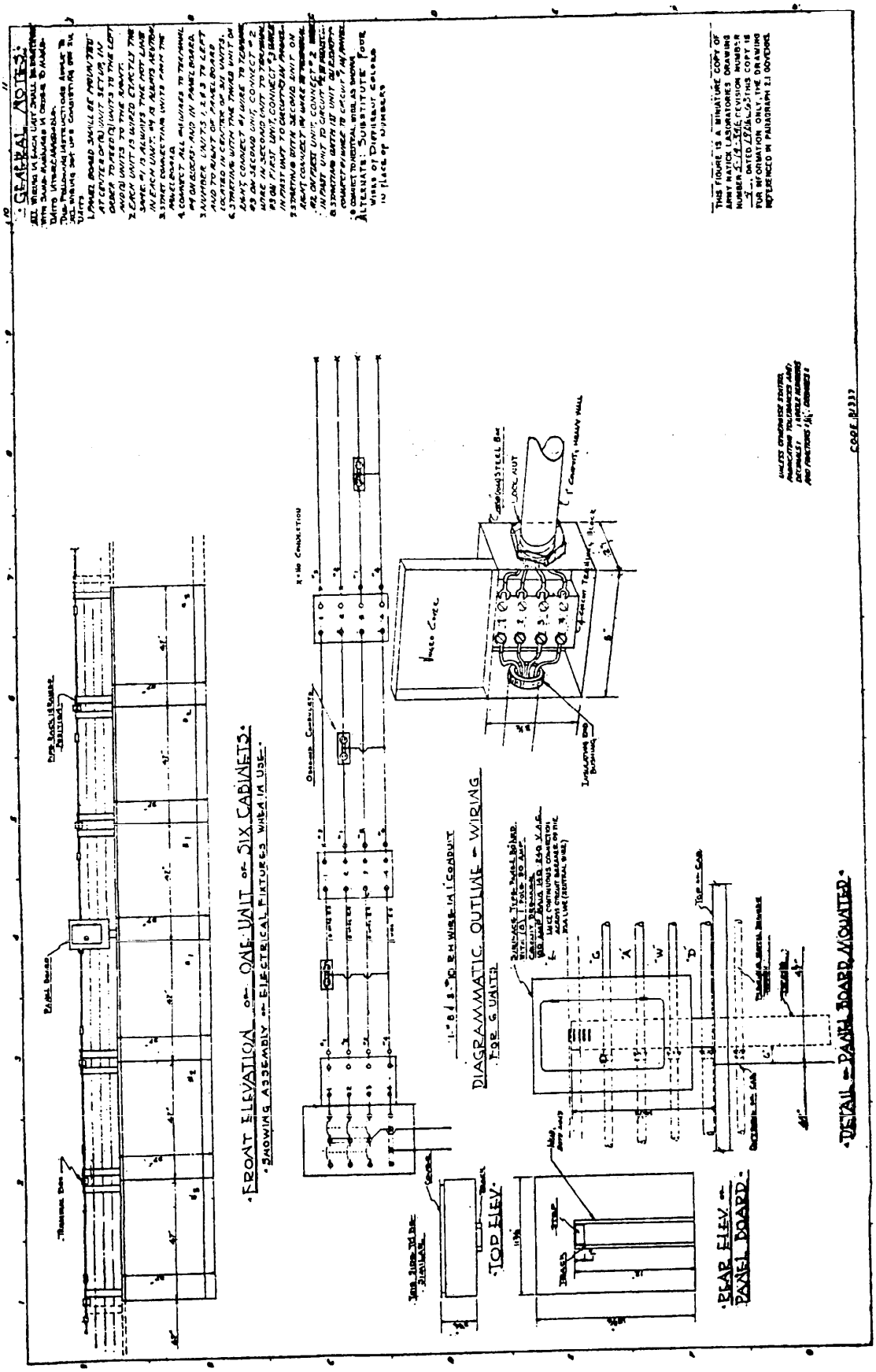


FIGURE 5. *Laboratory, petroleum, base; wiring diagram A.*

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